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Structural elucidation of the hexameric MmpS4-MmpL4 complex from *Mycobacterium tuberculosis*

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eLife Assessment

This study provides **valuable** insights through the elucidation of the first full-length structure of the heterohexameric (MmpS4)₃-(MmpL4)₃ transporter complex from *Mycobacterium tuberculosis*, advancing understanding of its transport mechanism, linked to virulence and drug resistance. The structural analysis is **convincing**, offering a clear framework for future mechanistic studies. Major strengths include a comprehensive structural characterization of the complex, though some conclusions require further validation.

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Abstract

Mycobacterium tuberculosis contains thirteen Mycobacterial membrane protein Large (MmpL) transporters, which belong to the family of secondary active RND transporters. MmpL4 and MmpL5, together with their operon partners MmpS4 and MmpS5, export the mycobacterial siderophore mycobactin and the last resort TB drug bedaquiline. Recently, we determined a structure of the MmpL4 monomer in complex with desferrated mycobactin, which lacked a functionally essential coiled-coil domain predicted to extend far into the periplasm. Here, we present a cryo-EM structure of the hexameric (MmpS4)₃-(MmpL4)₃ complex, which was enabled by rational disulfide cross-links based on AlphaFold predictions. We observed density for the coiled-coil domain, which protrudes into the periplasmic space at an angle of around 60° relative to the symmetry axis of the MmpL4 trimer. In the context of the hexameric complex, MmpL4's conformation differs strikingly from the one observed for monomeric MmpL4, which includes formation of a large cavity in the periplasmic domain and rearrangements of conserved proton coupling residues at the transmembrane domain. Our work provides an experimental workflow to obtain single particle cryo-EM structures of labile multiprotein complexes by AlphaFold-informed stabilization of predicted protein interfaces.

Introduction

Mycobacterium tuberculosis encodes 13 resistance-nodulation-cell division (RND) transporters, referred to as the Mycobacterial membrane protein Large (MmpL) family [1]. Multiple MmpL transporters are of interest as novel targets for anti-TB drugs. This includes the only essential member of the MmpL family, the trehalose monomycolate (TMM) exporter MmpL3 [2, 3]. Extensive characterization by X-ray crystallography and cryo-EM revealed the molecular details of how MmpL3 inhibitors and TMM substrates are bound [4–6] and clinical trials for small molecules targeting MmpL3 are ongoing [7]. Two additional MmpL transporters, the siderophore (mycobactin) exporters MmpL4 and MmpL5, are also of clinical interest. These transporters are

essential under low iron conditions and strongly contribute to virulence in mice [8, 9]. Importantly, MmpL4 and in particular MmpL5 confer resistance to bedaquiline, an antibiotic crucial for the treatment of multidrug resistant *M. tuberculosis* [10–13].

Like other RND transporters, MmpL proteins are proton-driven antiporters composed of a transmembrane domain and a periplasmic domain [2, 3]. The transmembrane domain consists of two pseudosymmetric six-helix bundles and contains conserved residues that are crucial for proton translocation. In case of MmpL3, MmpL4 and MmpL5, two DY-pairs located at the center of the transmembrane domain have been suggested to enable proton coupling [14–16]. The periplasmic domain is made up of two porter domains, referred to as the N-terminal (PN) and C-terminal (PC) porter domain. These two domains are expected to undergo rigid body movements that allow substrate transport through these periplasmic domains similar to the conformational changes observed in the *E. coli* multidrug efflux pump AcrB [17]. Structures of MmpL3 in which the PN and PC adopt an open conformation enclosing a large cavity and of MmpL4 in which the domains adopt a closed conformation provided initial support for this hypothesis [4, 15]. However, the mechanistic basis of how these conformational changes are coupled to protonation of the DY-pairs in the transmembrane domain remains elusive.

Many but not all MmpL proteins contain an additional coiled-coil domain (CCD) inserted in the PC, a structural element extending far into the periplasm. The CCD was first described based on the AlphaFold prediction of MmpL4, and was proposed to be involved in multimerization of MmpL proteins [15]. Based on the presence or absence of the CCD the MmpL family can be divided into two phylogenetic cluster. The CCD is the hallmark of cluster I MmpL transporters which contains most members of the family and differentiates it from cluster II MmpL transporters that lack this domain (MmpL3, MmpL11 and MmpL13). Additionally, only cluster I MmpLs are associated with a periplasmic accessory protein called Mycobacterial membrane protein small (MmpS), which is composed of a single transmembrane helix and a periplasmic β -sheet sandwich domain. The genes encoding five out of the ten cluster I MmpL transporters (*mmpL1*, *mmpL2*, *mmpL4*, *mmpL5*, *mmpL6*) form an operon with genes encoding their respective MmpS proteins.

Despite the biological and clinical importance of cluster I MmpL transporters, including MmpL4 and MmpL5, the first structures were determined only recently. *M. smegmatis* MSMEG_1382 was the first published structure of a cluster I MmpL transporter, but the CCD was not resolved due to the flexibility of the domain [18]. The second published cryo-EM structure was that of monomeric *M. tuberculosis* MmpL4 in complex with its substrate mycobactin, determined by our lab [15]. To obtain this structure the CCD had to be genetically truncated as full-length MmpL4 preparations consisted of multimeric assemblies that were not amenable for structure determination [15]. Using truncated constructs of MmpL4 and MmpL5 we also demonstrated that this domain is essential for mycobactin and bedaquiline export in *M. tuberculosis* using a mycobactin toxicity assay and bedaquiline susceptibility assay [15].

A role of the CCD in trimer formation was uncovered in a biochemical study of the cluster I *M. smegmatis* MmpL10 transporter (MSMEG_0410) [19]. Additionally, a contribution of the MmpS protein to MmpL trimer formation was described based on single-molecule TIRF microscopy measurements of GFP-tagged MmpL5 in mycobacterial cells [20]. Recent structures of hexameric *M. tuberculosis* MmpS5-MmpL5 and *M. smegmatis* MSMEG_1381-MSMEG_1382 complexes confirmed that both the CCD and MmpS protein contribute to MmpL trimerization [16, 21]. As predicted by AlphaFold, three CCDs interact extensively to form an elongated stalk. However, apart from these interactions few other contacts are made directly between the MmpL protomers. Instead, the MmpS proteins form extensive contacts with the transmembrane and porter domains of multiple MmpL protomers and the MmpS β -sheet sandwich domains surround the CCD stalk [16, 22]. Even though both the CCD and MmpS contribute to MmpL trimerization, neither are strictly required to obtain a stable multimeric assembly of the MmpL complex. This was demonstrated by a structure of hexameric *M. tuberculosis* MmpS5-MmpL5 wherein the CCD was genetically deleted [22] and purification of trimers of full-length MmpL5 in the absence of MmpS5 [16].

MmpL4, MmpL5 and MSMEG_1382 co-purify with abundant acyl-carrier proteins (ACPs) from the respective expression host, namely *E. coli* ACP in case of MmpL4 [15] and AcpM in case of MmpL5 [16] and MSMEG_1382 [23]. Biochemical analysis showed that MmpL4 also interacts with MbtL (Rv1344), an ACP involved in mycobactin biosynthesis, suggesting that mycobactin biosynthesis and export are coupled processes [15].

In this work, we utilized the AlphaFold2 model of (MmpS4)₃-(MmpL4)₃ to rationally design disulfide bonds between MmpS4 and MmpL4. These engineered disulfide cross-links stabilized the MmpS4-MmpL4 interface, thereby enabling cryo-EM structure determination of hexameric MmpS4-MmpL4. This structure revealed a binding cavity in the periplasmic domain of MmpL4 and a tilted conformation of the CCDs.

Results

Prediction of a hexameric MmpS4-MmpL4 complex using AlphaFold2

In order to purify MmpL4 from *E. coli*, *mmpL4* had to be expressed together with its operon partner *mmpS4* [15]. When full-length MmpL4 (including its CCD) was purified via a C-terminal GFP-tag, the sample predominantly eluted as a broad peak from the size exclusion chromatography (SEC) column, corresponding to MmpL4 multimers of varying stoichiometries [15]. A later-eluting peak contained the MmpL4 monomer and suffered from protein degradation. SDS-PAGE analysis and mass-spectrometry analyses of the early-eluting peak revealed that MmpS4 was co-purified, as well as the acyl-carrier protein ACP of the expression host *E. coli*. However, when attempting to determine a cryo-EM structure of the multimeric MmpS4-MmpL4 complex, the sample quality was poor, only permitting reconstruction of a 6.3 Å map of the MmpL4 monomer in complex with ACP [15]. Hence, in our hands, the detergent-purified MmpS4-MmpL4 complex was not stable enough to be structurally investigated.

Therefore, we used AlphaFold2 to predict the multimeric complex including three MmpL4 and three MmpS4 chains [24]. AlphaFold2 generated a model with high values for the predicted local distance difference test (pLDDT) score and importantly also for the predicted alignment error (PAE) over the entire polypeptide chains of MmpL4 and MmpS4 indicating a high confidence in the predicted interactions (Fig. 1A, Fig. S1). The most characteristic hallmark of the (MmpS4)₃-(MmpL4)₃ model is the CCD of MmpL4 which extends along the three-fold symmetry axis 130 Å far into the periplasm. MmpS4 is predicted to be anchored in the cytoplasmic membrane via an N-terminal single transmembrane helix. If we follow the polypeptide sequence of one MmpS4 protomer in the AlphaFold2 model, the transmembrane helix of MmpS4 associates with the 12-transmembrane helix bundle of the first MmpL4 protomer (petrol in Fig. 1A) by interacting with transmembrane helix 8. A linker devoid of secondary structures bridges the crevice formed between the first (petrol) and second (white) MmpL4 protomer, thereby establishing interactions with both MmpL4 protomers. The C-terminal domain of MmpS4 features the topology of a β-sheet sandwich [8, 25], wherein two β-sheets each comprising four β-strands each stack against each other. This architecture is further reinforced by a disulfide bond connecting that connects the sheets. The MmpS4 β-sheet sandwich establishes extensive contacts with the CCD of MmpL4, especially the α-helical hairpins donated from the second (white) and the third (grey) MmpL4 protomer. In summary, MmpS4 is predicted to establish extensive contacts with all three MmpL4 protomers, thereby stabilizing the MmpL4 trimer, and in particular the CCD.

Experimental validation of the AlphaFold2-predicted MmpS4-MmpL4 complex

We used targeted disulfide cross-linking to experimentally validate the AlphaFold2-predicted MmpS4-MmpL4 complex. Spontaneous disulfide bonds are expected to form in the oxidative environment of the periplasm of *E. coli* if cysteine pairs are sufficiently close to each other, especially if they are placed in flexible loop regions. The same experimental strategy was

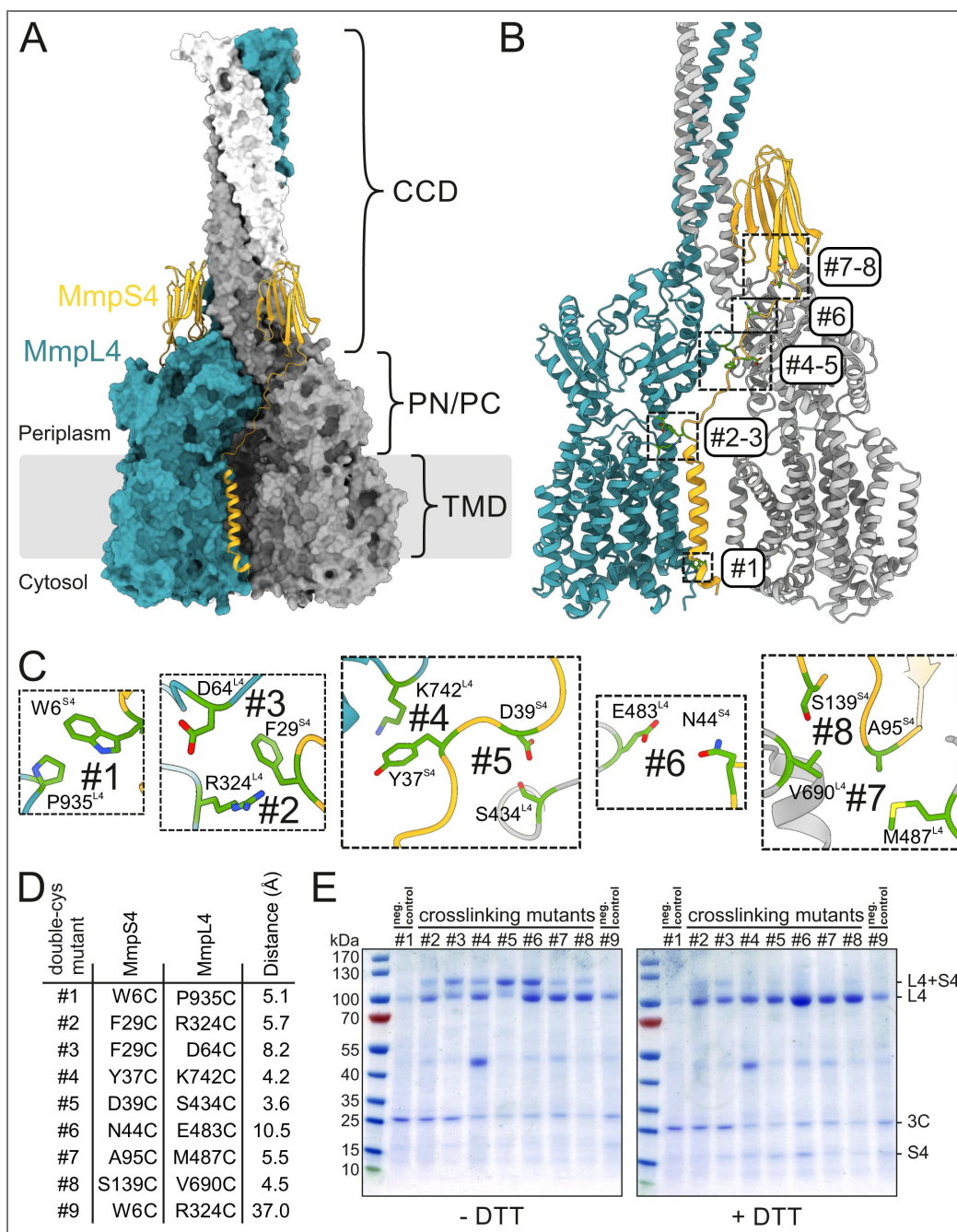


Figure 1. Engineered MmpS4-MmpL4 disulfide cross-links based on AlphaFold2 prediction.

(A) AlphaFold2 prediction of the hexameric (MmpS4)₃-(MmpL4)₃ complex. The three protomers of MmpL4 are shown in surface presentation and colored petrol, grey and white. The MmpS4 protomers are shown as yellow cartoon. The boundary of the cytoplasmic membrane is indicated by grey lines. The locations of the transmembrane domains (TMD) and the periplasmic PN/PC domains are indicated. MmpL4 is predicted to form a large coiled-coil domain (CCD) extending into the periplasm. (B and C) Based on the AlphaFold2 model, eight double-cysteine mutants (#1-#8) were generated, which exhibit a close distance between MmpL4 and MmpS4. The original residues that were substituted by cysteines are shown as sticks. The mutants #2-#8 are expected to form spontaneous disulfide cross-links in the oxidative environment of the periplasm. (D) List of characterized double cysteine mutants. Whenever possible, the distances were measured between the γ -atoms of the respective residues of MmpS4 and MmpL4. In case of alanine and proline residues, the atom nearest to the γ -atom of the partner residue was used to measure the distance. (E) Non-reducing (left) or reducing (right) SDS-PAGE analysis of purified cysteine cross-linking mutants of MmpS4-MmpL4, stained by coomassie. The negative control cross-linking pair #1 was not expected to form a disulfide bond because it is located in the reducing environment of the cytosol. The negative control cross-linking pair #9 bears cysteines that are too far apart to form a disulfide bond.

previously utilized to study the conformational cycling of the RND transporter AcrB [26], but also more recently to validate the MmpS5-MmpL5 complex structure [16]. To avoid unspecific reactions with free thiol groups, we mutated five out of the seven naturally occurring cysteines of MmpL4 into serines (see methods). We decided to keep two highly conserved cysteine-pairs that form disulfide bonds, the first one at the tip of the CCD of MmpL4 and the second one between the β -sheets of MmpS4. The resulting cys-depleted MmpS4-MmpL4 construct behaved similarly to the original progenitor in terms of protein yield and quality (Fig. S2 [4]). In addition, cys-depleted MmpS4-MmpL4 was fully functional in our mycobactin toxicity assay (Fig. S3 [4]).

Based on the AlphaFold2 prediction we introduced eight cysteine pairs (numbered #1-#8) with close distances between the thiol group into flexible loops along the MmpS4-MmpL4 interface (Fig. 1B and C [4]). Cysteine pair #1 located at the cytosol served as a negative control, because it is not expected to form spontaneous disulfide bonds due to the reducing environment. Cysteine pair #9 was included as an additional negative control, because its thiols are 37 Å away from each other and therefore cannot form disulfide bonds (Fig. 1D [4]). The respective proteins were produced in *E. coli*. Prior to the detergent solubilization step, *n*-ethylmaleimide was added to react with free reactive thiols and prevent undesired disulfide cross-links between transporter complexes during detergent extraction and subsequent purification steps. Subsequently, the transporter complexes were purified via GFP attached C-terminally to MmpL4 and analyzed by non-reducing and reducing SDS-PAGE (Fig. 1E [4]). Cysteine pairs #2-8 established variable degrees of MmpS4-MmpL4 cross-linking. As expected, the cytoplasmic pair #1 and the negative control pair #9 did not cross-link, thereby validating the experimental approach. The degree of cross-linked protein for cysteine pair #5 (D39C^{MmpS4}-S434C^{MmpL4}), which according to the AlphaFold2 prediction has the closest distance of 3.6 Å between the thiol moieties, was the highest. In summary, the engineered cysteine cross-links provided experimental evidence for the interaction between and trimerization of MmpS4 and MmpL4 as predicted by the AlphaFold2 model.

Cryo-EM structure of the (MmpS4)₃-(MmpL4)₃ complex

The cysteine cross-linking pair #5, which exhibited the strongest cross-linking efficiency, was purified (Fig. S2 [4]) and analyzed by single particle cryo-EM, resulting in a high-resolution reconstruction of the hexameric (MmpS4)₃-(MmpL4)₃ complex (Fig. 2 [4], Fig. S4 [4], Fig. S5 [4], Table S1 [4]). We obtained a 3.1 Å map without applying symmetry during data processing (Fig. S4 [4]). An additional density for the *E. coli* ACP was present at the cytosolic interface of the transmembrane domain, similar to what has previously been described in the structure of the MmpL4 monomer. Because this density was only weakly defined (Fig. 2A [4], Fig. S4 [4]), we did not include it in our model. The C1 cryo-EM map contained well-defined density for the transmembrane helices and part of the flexible linkers of MmpS4, but not for its β -sheet sandwich domains (Fig. 2A [4], Fig. S5 [4]). Well-resolved density is present for the transmembrane domains and the periplasmic PN and PC domains of MmpL4, whereas the CCD is less well defined (Fig. 2A [4] and B [4]). Strikingly, the CCD is tilted by around 60° away from the threefold symmetry axis rather than extending along it as predicted by AlphaFold2. Only two out of three α -helical hairpins of the three MmpL4 protomers are resolved (Fig. 2C [4]), which may explain its tilted conformation. Using a local refinement approach with a mask containing the CCD and the uppermost regions of MmpL4, the cryo-EM density of the CCD and the loops connecting to the PC domain of MmpL4 was further improved (Fig. S4 [4] and Fig. S5 [4]), enabling us to build a model of the lower part of the CCD, including the linkers connecting to the PC of MmpL4. Model building revealed that the tilted CCD is composed of the α -helical hairpins of the polypeptide chains A and B. To assess whether there are asymmetries in other parts of the structure, we superimposed the individual protomers of the (MmpS4)₃-(MmpL4)₃ complex analyzed using C1 symmetry (Fig. 2E [4]). Apart from the two resolved α -helical hairpins, the MmpL4 core domains and the resolved parts of MmpS4 differ by a RMSD of less than 0.6 Å and are therefore structurally identical considering the map resolution of around 3 Å. The fact that the core domains of MmpS4 and MmpL4 do not deviate between the protomers argues against the possibility that the cross-links established between them cause the (asymmetric) tilt of the CCD. To increase the resolution of the core parts of the (MmpS4)₃-

(MmpL4)₃ complex, we applied C3 symmetry resulting in a map with a resolution of 2.9 Å (Fig. S4). Using this data, we built a model comprising residues E16-M487 and V690-R939 of MmpL4 and residues T5-T40 of MmpS4.

Identification of a deep cavity accommodating a detergent molecule

A systematic inspection of non-proteinaceous densities revealed a strong detergent-like density in a cavity formed between the PN and PC domains and the upper part of the TMD (Fig. 3A and B). We modeled the density as a N-dodecyl β-D-maltoside (β-DDM) molecule, the detergent used to extract and purify the transporter complex. The two sugar moieties are sandwiched between the PN and PC and the twelve carbon atoms of the aliphatic chain are accommodated in a hydrophobic patch in the transmembrane domain (Fig. 3B). We did not observe density for β-DDM at this location in the monomeric MmpL4 structure, although β-DDM was also used for protein extraction and purification. Using the 3 V tool [27], we analyzed the cavity that accommodates β-DDM in the (MmpS4)₃-(MmpL4)₃ complex and found that it spans from the TMD into the interface between PN and PC (Fig. 3D). The cavity does not open up to the periplasm as observed for the open conformation of MmpL3 (Fig. 3E). Instead, an entrance is found between transmembrane helices 1 and 2, at the same position at which Fountain et al. observed a prominent non-proteinaceous density in the MmpS5-MmpL5 complex structure [22]. In striking contrast, monomeric MmpL4 only contains a very small, occluded cavity at the same location (Fig. 3C). We also noted extra-densities wedged between the transmembrane helix bundles at the inner leaflet of the membrane, in analogy to the lipids observed in the MmpS5-MmpL5 structures that were suggested to stabilize the MmpL5 trimer [16, 22]. Because they were not sufficiently defined to unambiguously differentiate between lipid or detergent molecules we refrained from interpreting them. It should be noted that we purified the MmpS4-MmpL4 complex from *E. coli* as expression host, whereas the MmpS5-MmpL5 complexes were both purified from *M. smegmatis*.

Comparison of monomeric and trimeric MmpL4 structures

To systematically analyze the conformational differences between monomeric and multimeric MmpL4, we used the Needleman–Wunsch algorithm implemented in ChimeraX to superimpose the 3.0 Å MmpL4 monomer structure (PDB ID: 9GIO) onto the first MmpL4 protomer (chain A) of the asymmetric (MmpS4)₃-(MmpL4)₃ complex structure determined at 3.1 Å (Fig. 4A). The RMSD was 2.14 Å over 722 residues, demonstrating that MmpL4 exhibits substantial structural differences depending on whether it is a monomeric protein or embedded in the hexameric complex. When plotting the Ca-atom distance of monomeric versus trimeric MmpL4 (Fig. 4C), we found that conformational differences affect all parts of the transporter. Most prominently, we observed a rotation of the PN domain (Fig. 4C), which is responsible for creating the deep cavity accommodating β-DDM in the complex structure (Fig. 3A and B). Intriguingly, we also observed a rearrangement of the conserved DY-pairs, four key residues residing at the center of the transmembrane domain that are suggested to couple active transport to the proton-gradient [15] (Fig. 3C and D, Fig. S6). While Y273 and D844 are around 5 Å apart in monomeric MmpL4 (Fig. 3C), they engage in a hydrogen bond in the context of the hexameric complex (Fig. 3D) in a similar configuration as observed for MmpL3 (PDB ID: 7NVH) (Fig. 3E). These rearrangements may be coupled to protonation events at D844 in a similar manner as characterized for the proton translocating aspartates in the drug efflux pump AcrB [28]. Concomitant with the subtle rearrangements in the proton-coupling DY-pairs, the twelve transmembrane helices of MmpL4 change their position (RMSD of 1.78 Å over 369 residues) to a similar extent as the conformational changes observed for the TMD of asymmetric AcrB [29]. In MmpL4, the conformational changes mostly affect the upper parts of the transmembrane helices, in particular transmembrane helix 2-4 and 11-12 (Fig. 4B and C). These conformational changes result in a larger distance between transmembrane helices 1 and 2, creating the channel

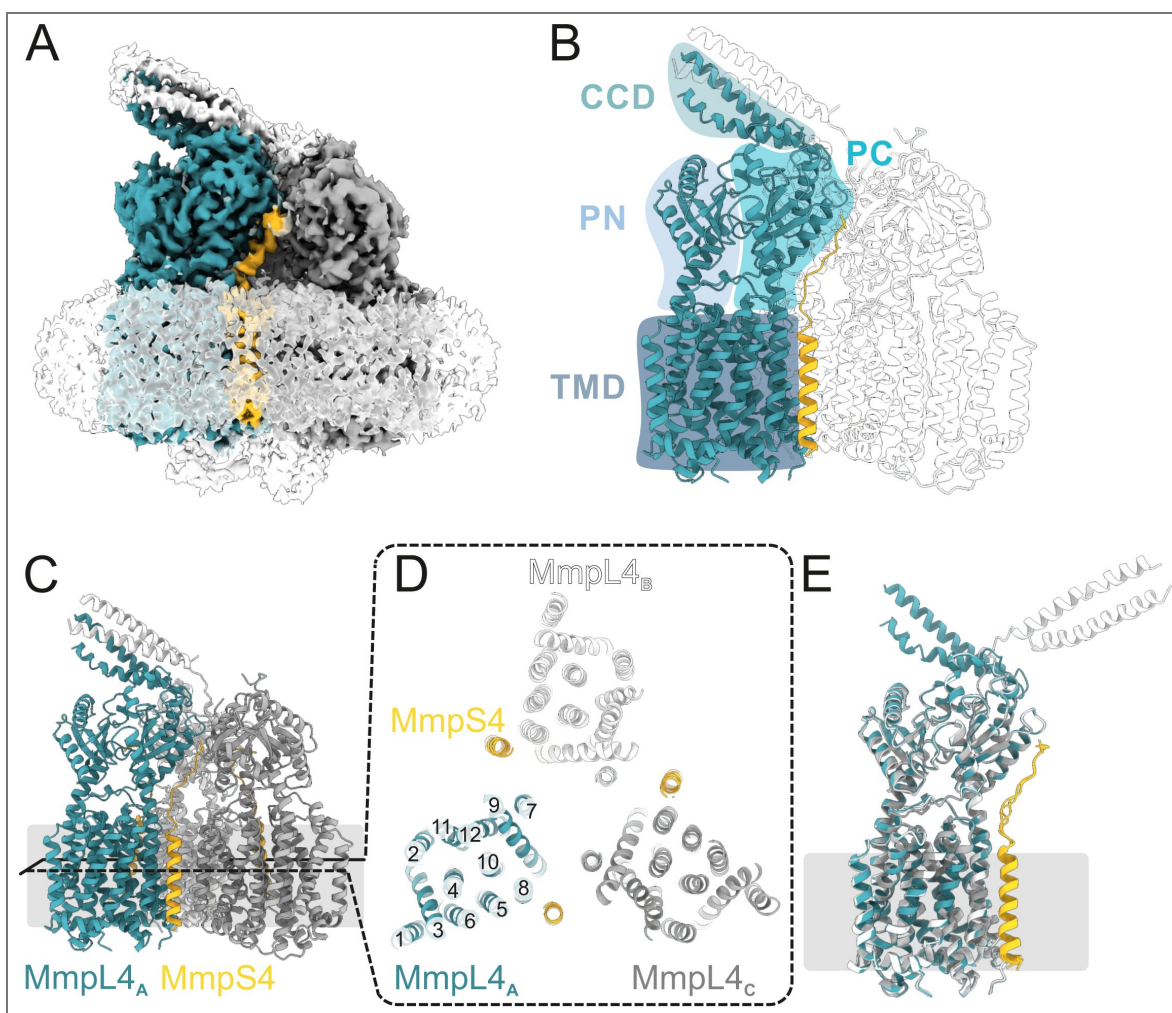


Figure 2. Structure of the hexameric (MmpS4)₃-(MmpL4)₃ complex.

(A) Left panel; Cryo-EM map of the hexameric (MmpS4)₃-(MmpL4)₃ complex processed using C1 symmetry viewed along the membrane plane and contoured at 4σ. The detergent micelle and weak density for *E. coli* ACP are shown as white, transparent density. MmpL4 protomers are colored in petrol, grey and white, MmpS4 in yellow. Density for a tilted CCD formed out of two α-helical hairpins donated from MmpL4 protomers A (petrol) and B (white) is visible. (B) Schematic labeling of MmpL4 domains. (C) Structural model of the asymmetric (MmpS4)₃-(MmpL4)₃ complex shown as cartoon shown along the membrane plane (left). (D) Transmembrane helices of MmpL4 and MmpS4 viewed from the periplasm. The transmembrane helices of MmpL4 are numbered. (E) Superimposed MmpS4-MmpL4 monomers of asymmetrically processed (MmpS4)₃-(MmpL4)₃ asymmetric orientation of the α-helical hairpins. complex, showing the asymmetric orientation of the α-helical hairpins.

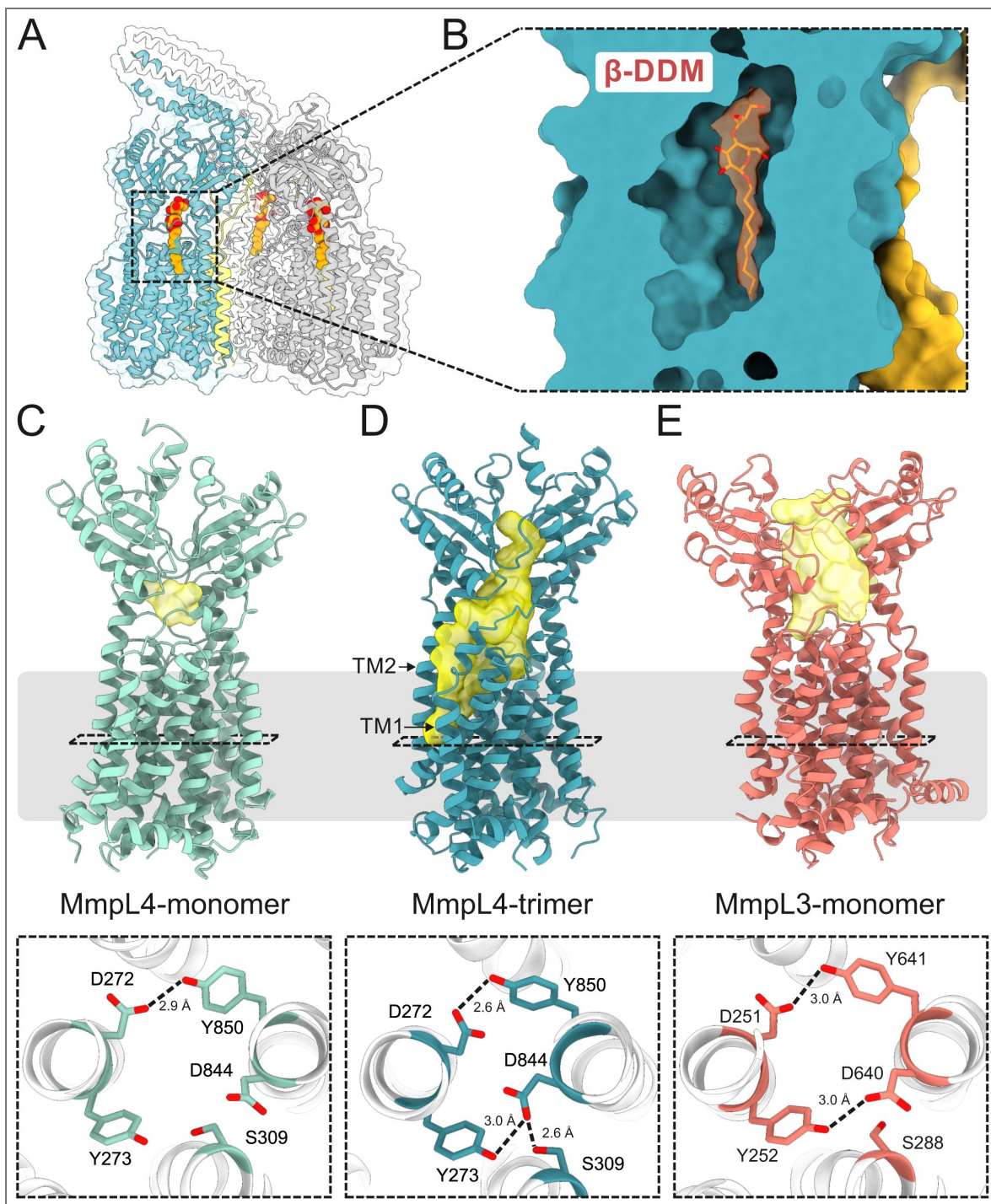


Figure 3. Detergent binding in a deep cavity extending into the periplasmic domain of MmpL4.

(A) Structural model of hexameric (MmpS4)₃-(MmpL4)₃ complex processed using C1 symmetry. Three detergent molecules (β -DDM) bound to each of the MmpL4 protomers are depicted as red/orange spheres. (B) Enlarged view of the β -DDM binding site of chain A. Non-proteinaceous density (contoured at 6.8 σ) interpreted as β -DDM is depicted in orange. (C – E) Cavity analysis of monomeric MmpL4 (PDB ID: 9GI0) (C), trimeric MmpL4 (this work, PDB ID: 9SYJ) (D) and monomeric MmpL3 (PDB ID: 7NVH) (E) using the 3V tool. Cavities are shown in yellow and the membrane boundary is indicated. The bottom row shows the conserved DY-pairs and S309(MmpL4)/S288(MmpL3) viewed from the periplasm. Hydrogen bonds are indicated by dashed lines.

entry that leads to the periplasmic cavity (Fig. 3D). It remains to be shown in future work whether the structural differences observed between monomeric and multimeric MmpL4 represent alternate conformational states that are relevant for efflux function.

Structural features of MmpL CCDs analyzed by AlphaFold

The presence of defined density for the tilted CCD consisting of α -helical hairpins of only two MmpL4 protomers suggests that it represents a thermodynamically stable intermediate. The density for the CCD was less well defined than other parts of MmpL4, suggesting that the CCD is rather flexible (Fig. S5). To gain further insights into this unique structural element, we used AlphaFold3 to predict the CCDs of all cluster I MmpL proteins of *M. tuberculosis* (Fig. S7A) and performed a sequence analysis of this region (Fig. S7B). In terms of phylogeny, Cluster I MmpL proteins can be further subdivided into two groups, the first one encompassing MmpL1, MmpL2, MmpL4, MmpL5, MmpL6 and MmpL9 (which apart from MmpL9 are encoded in an operon with a respective *mmpS* gene) and the second one encompassing MmpL8, MmpL10 and MmpL12 (Fig. S7). MmpL7 cannot be clearly assigned to either of the groups. The predicted CCDs differ quite strikingly between the two phylogenetic groups, with the second group (MmpL8, MmpL10 and MmpL12) being substantially longer and containing kinks and discontinuities in the α -helices. It should be noted that the AlphaFold3 pLDDT values were less confident for MmpL7, MmpL8, MmpL10 and MmpL12 as compared to the other MmpLs, and future experimental structures will be required to validate their correctness.

The most notable feature of the CCDs is the enrichment of methionine side chains pointing towards the enclosed channel for many, but not all MmpL proteins (Fig. 5A, Fig. S7). The channel wall of MmpL4's CCD, for example, is lined with 45 methionines (15 per α -hairpin). By contrast, MmpL6 has one methionine per α -hairpin, suggesting that the methionines are not *per se* needed for the assembly of the CCD, but instead likely play a functional role during transport.

The CCD channel radius of all MmpL proteins appears to be very narrow. The MOLEonline tool [30] predicts several constrictions and a minimal channel radius of 1.1 Å for the CCD of MmpL4 (Fig. 5B). It should be kept in mind that these analyses were performed with AlphaFold predictions and not with experimental structures. In its proposed closed conformation, the channel is too narrow to let bulky substrates such as desferrated mycobactin or bedaquiline pass through, even if one considers flexibility of long methionine side chains. Further experimental studies will be required to deepen our mechanistic understanding of the CCD, in particular with regards to its role in substrate transport.

Discussion

Structure predictions using AlphaFold have become a standard step in every structural biology project, due to its ease of use and the high accuracy of its predictions [24]. Predictions of multi-protein complexes by AlphaFold2 multimer (and more recently AlphaFold3) are especially attractive, because they provide potential cues about the interaction interfaces as well as possible multimeric assemblies [31, 32]. In the context of MmpL and MmpS proteins, AlphaFold has been extensively used by us and others to firstly identify the CCD in MmpL4 and MmpL5 and to secondly predict its involvement in trimerization of the proteins [15, 16, 19, 22]. In our previous study on MmpL4, we cloned a transporter construct devoid of the predicted CCD to enable the determination of a highly resolved cryo-EM structure of the MmpL4 monomer [15]. An analogous strategy was also chosen by Fountain et al. to obtain the multimeric MmpS5-MmpL5 structure [22]. In this study, we used the hexameric structure of MmpS4-MmpL4 predicted by AlphaFold2 as the basis to introduce engineered cysteine cross-links. This strategy allowed us to biochemically validate the predicted model and facilitated its structural determination. The procedure described in this work may serve as a blueprint on how AlphaFold predictions can aid the rational stabilization of difficult-to-handle protein complexes. Other groups have determined cryo-EM complex structures of the MmpS4-MmpL4 homologues MmpS5-MmpL5 [16, 22] and MSEM_1381-1382 [21]. These structures were determined without cross-linking, either by purifying the complexes from *M. smegmatis* [16, 22] (instead of *E. coli* as in this study) or by using grid

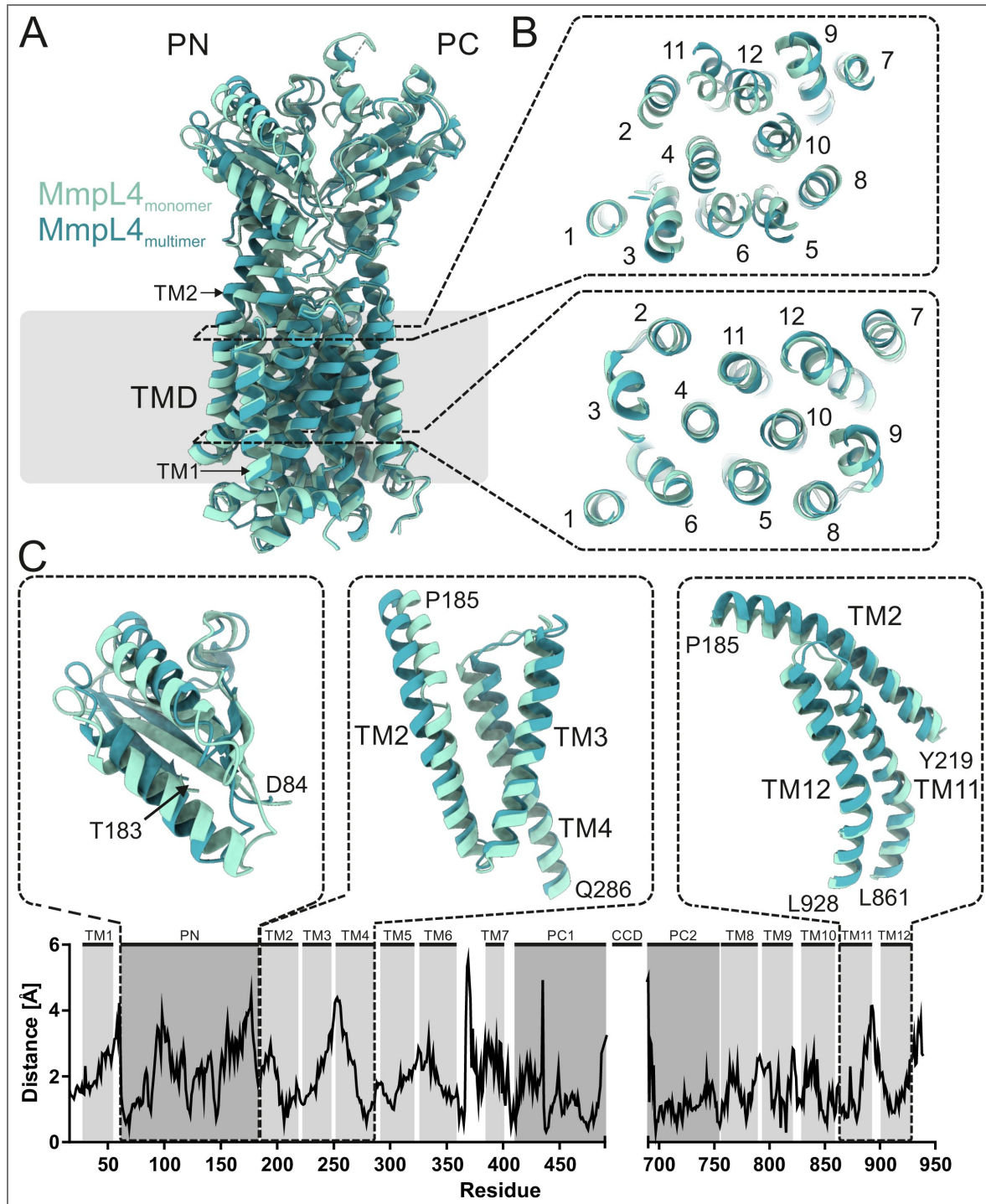


Figure 4. Systematic structural comparison of monomeric versus trimeric MmpL4.

(A and B) Previously determined monomeric MmpL4 colored in light blue (PDB ID: 9GI0) was superimposed onto trimeric MmpL4 colored in dark blue (chain A of structure obtained using C1 symmetry, PDB ID: 9SYJ). (B) Two slices along the membrane plane are shown from the periplasm onto the TMDs at the two heights indicated by the dashed rectangles in (A). (C) C α -atom distance plot between monomeric and trimeric MmpL4. The respective TM helices and domains are indicated above. For three structural elements showing marked differences, the superimposed structures are shown as cartoon above the plot. No structural information was available for the CCD and the C-terminus. The overall RMSD is 2.14 Å over 722 residues.

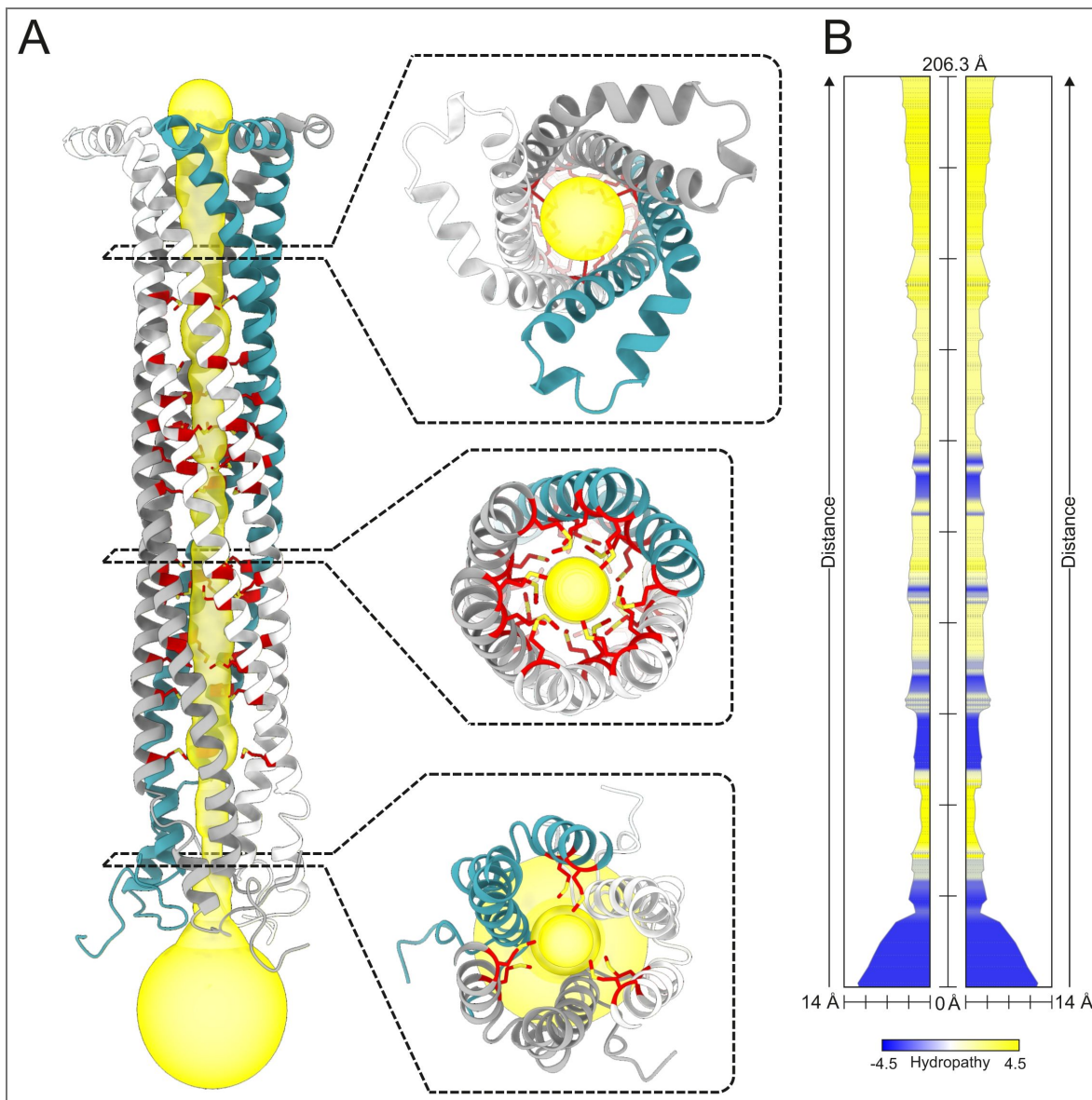


Figure 5. Channel prediction of MmpL4's CCD.

(A) The CCD of trimeric MmpL4 as predicted by AlphaFold3 and shown as cartoon. A continuous channel leading through the CCD interior was calculated by the MOLEonline tool and shown as yellow surface. Methionine side chains lining the channel surface are shown as red sticks. Three top views along the pore are shown as insets. (B) Radius and biophysical properties of the channel shown in (A) calculated by the MOLEonline tool.

preparation methods that avoid purification of the membrane protein complexes after solubilization [21]. In both approaches, hexameric complexes composed of three MmpS and three MmpL proteins were observed, demonstrating that cross-linking via engineered cysteines did not result in structural artefacts. In addition, the MmpS43-MmpL43 complex we obtained is structurally very similar to the MmpL5 structures obtained by the other two groups (RMSD values smaller than 1.4 Å for the core domains of MmpL4/MmpL5) (Fig. S8 [21]).

AlphaFold predictions and structures of the multimeric assembly of MmpS-MmpL proteins presented in this study and by others [16, 21, 22] provided important insights into the architecture of these unique efflux machineries. The structures established the role of the MmpS protein in stabilizing the MmpL trimer by interactions of every MmpS protomer with all three MmpL protomers. We did not observe density for the β -sheet sandwich domains of MmpS4, nor were the respective domains of MmpS5 observed in the symmetric hexameric MmpS5-MmpL5 complex [16]. Our cross-linking data, however, strongly suggests that the β -sheet sandwich domains of MmpS4 interact with the CCD in the configuration predicted by AlphaFold2 (see cross-links of cysteine pairs #7 and #8 in Fig. 2A [21]) and this was confirmed by the cryo-EM structure of MSEM_1381-1382 [21]. We therefore hypothesize that the role of the β -sheet sandwich domains of MmpS proteins is to further stabilize the fully assembled CCD and orchestrate the orientation of the CCD in the periplasm.

The CCDs of cluster I MmpL proteins are particularly interesting from a structural and functional point of view. In their fully assembled, upright and trimeric conformation, the CCDs protrude into the periplasm, as noted in our previous MmpL4 study [15] and also suggested by other research groups [16, 21, 22]. A length estimate based on the cryo-ET analysis of the mycobacterial cell membranes suggests that the upright CCD does not bridge the entire periplasmic space of *M. tuberculosis* [22], suggesting that further protein partners are needed to complete the bridge. In the hexameric (MmpS4)3-(MmpL4)3 complex structure presented in this work, the CCD adopts a partially assembled, tilted conformation, consisting of only two α -helical hairpins. A general finding from our and other studies is the overall flexibility of the CCD. In our structure, the density for the tilted CCD is less defined than other parts of the transporter (Fig. S5 [21]). Similarly, the CCD of MmpL5 was poorly defined in the MmpS5-MmpL5 complex structure determined with full-length MmpL5 [16], while in the study of Fountain et al., the CCD of MmpL5 was genetically truncated [22]. In contrast, the MSEM_1381-1382 complex structure revealed a fully assembled, upright CCD and a clear density for the β -sheet sandwich domains of the MmpS protein intimately interacting with the CCDs, as predicted by AlphaFold2 [21], which may be explained by a more robust CCD of MSEM_1382 compared to MmpL4 and MmpL5 or additional stabilization by the β -sheet sandwich domains of MSEM_1381.

The channel radius of the MmpL4 CCD is very narrow (minimum of 1.1 Å) according to the AlphaFold3 prediction (Fig. 5 [21]). In addition, the cryo-EM structure of MSEM_1382 revealed a constriction in the CCD channel [21]. Even if the methionine side chains lining the channel wall are considered to be flexible, large conformational changes of the α -helical hairpins relative to each other would be required to allow passage of molecules as large as mycobactin and bedaquiline. In the context of the AcrAB-TolC efflux machinery, iris-like opening and closing movements have been described for the outer membrane component TolC [33], and similar helical twisting may widen the channel of the CCD. Alternatively, one might speculate that the substrates are transported along the CCD surface, potentially requiring further protein partners. It is interesting to note that siderophore secretion and drug efflux by MmpL4/MmpL5 systems involves at least two additional proteins, namely the periplasmic protein Rv0455, which was shown to be essential for mycobactin efflux [34] and an outer membrane channel, whose identity remain elusive. A complete molecular understanding of the transport mechanism through the MmpL4/MmpL5 systems hence requires the identification of the missing components and structural information about their interactions.

Finally, the route of substrate transport through the transmembrane domain and periplasmic domains of MmpL4 and MmpL5 remains to be further investigated. In our first study on MmpL4, we identified a mycobactin binding site that is accessible from the cytosol [15]. Fountain et al.

recently performed extensive mutagenesis and gain-of-function experiments on MmpL5 and thereby identified several residues in the periplasmic domains involved in bedaquiline recognition and transport [22]. While the authors did not find a bedaquiline binding site, they identified a cleft between TM1 and TM2 which is larger in the multimeric assembly versus the MmpL5 monomer and contains non-proteinaceous density of unknown identity [22]. Similarly, in the context of the MmpS4-MmpL4 complex, we observed a large cavity occupied by β -DDM in the periplasmic domain, which has its entrance between TM1 and TM2. As has been learned for the multidrug efflux pump AcrAB-TolC through more than two decades of work, the identification of substrate transport routes is complex and not all substrates are transported through the same channels [35–37]. Interestingly, a subset of substrates including β -lactams, fusidic acid and β -DDM have been proposed to use a channel extending from a groove formed by TM1 and TM2 to reach their binding pocket [37]. A similar route may be taken by mycobactin and bedaquiline, though it has to be considered that AcrB contains a periplasmic domain architecture that is strikingly different from those of MmpL proteins.

Common to all transport routes through AcrAB-TolC is the necessity of the AcrB protomers (corresponding to MmpL4 and MmpL5 in our system) to undergo large conformational transitions resulting in the opening and closure of transport channels in the periplasmic domains to drive transport [17, 26, 38]. As a consequence, homotrimeric AcrB assumes an asymmetric conformation with pronounced structural deviations between the protomers, that reflect different protonation states of the key residues in the TMD responsible for coupling proton translocation to active transport [17, 28]. Apart from the tilted CCDs, the MmpL4 trimer in the multimeric structure presented in this work is highly symmetric (Fig. 4C). Likewise, no notable asymmetries were reported for the MmpL5 and the MSEM1382 trimers [16, 21, 22]. In contrast, we noted pronounced conformational differences of MmpL4 in its monomeric versus its trimeric context which are accompanied by conformational changes to the proton-translocating DY-pair residues. Further studies will be required to fully uncover the conformational cycle of MmpL proteins in their multimeric assemblies, which will be crucial to understand the mechanism(s) by which these unique multi-subunit efflux systems export mycobactin as well as clinically important TB drugs.

Methods

Strains, media and antibiotics

The *Escherichia coli* K-12 strain MC1061 was used for cloning and heterologous protein expression. Luria Broth (LB) liquid cultures and agar were used for cloning and Terrific broth (TB) liquid culture for protein expression. In *E. coli*, 100 $\mu\text{g ml}^{-1}$ ampicillin and 200 $\mu\text{g ml}^{-1}$ hygromycin were added to the medium when required. *M. tuberculosis* strains were grown at 37 °C in Middlebrook 7H9 liquid medium (Difco Laboratories) supplemented with 0.2% glycerol, 10% ADS, 0.2% casamino acids, 24 $\mu\text{g/ml}$ pantothenate, or 0.02% tyloxapol and 20 μM hemin. Hartmans de Bont (HdB) minimal medium (500 μM MgCl_2 , 7 μM CaCl_2 , 1 μM NaMoO_4 , 2 μM CoCl_2 , 6 μM MnCl_2 , 7 μM ZnSO_4 , 1 μM CuSO_4 , 15 mM $(\text{NH}_4)_2\text{SO}_4$, 12 mM KH_2PO_4 , pH 6.8) containing 1% (v/v) glycerol was supplemented with 0.2% casamino acids, 24 $\mu\text{g/ml}$ pantothenate, 0.02% tyloxapol and 20 μM hemin and was used for the Alamar Blue assays. Hygromycin was added at a concentration of 50 $\mu\text{g ml}^{-1}$ if needed.

Model predictions using AlphaFold2 and AlphaFold3 and channel predictions

The structure of the MmpS4-MmpL4 trimeric complex was predicted using AlphaFold2 (Parafold setup) in multimer mode [24, 39]. The structures of CCDs of MmpL proteins were predicted using the web interface of AlphaFold3 [40] by first predicting the respective hexameric MmpS-MmpL (in case a MmpS protein is part of the complex) or trimeric MmpL complexes. For simplicity, we only show the CCDs of these structures. The channel properties of the predicted CCD of MmpL4 were analyzed using the MOLEonline tool [30].

Cloning of mmpS4-mmpL4 cross-linking variants

The gene for *mmpS4-mmpL4-cys-depleted* (wherein C39, C319, C335, C393 and C780 of MmpL4 are substituted by serines) was synthesized by Twist Biosciences into the fragment exchange (FX) cloning compatible entry vector pINIT (Addgene #46858) [41]. The last 123 nucleotides of *mmpL4* were codon optimized for synthesis due to their high GC content. The full nucleotide sequence of *cys-depleted mmpS4/mmpL4* is shown in Table S3. The *mmpS4-mmpL4-cys-depleted* gene was subcloned into the expression vector pBXC3GH (Addgene #47070) by FX cloning [42]. Nine cysteine pairs were introduced into pBXC3GH_ *mmpS4-mmpL4-cys-depleted* following the QuikChange site-directed mutagenesis protocol using the primers listed in Table S2. For example, for the W6C^{MmpS4}-P935C^{MmpL4} cysteine pair first the W6C^{MmpS4} mutation was introduced to *mmpS4* using the primers mmpS4_W6_Fw and mmpS4_W6_Rv. After confirmation by Sanger sequencing, the P935C^{MmpL4} mutation was introduced into *mmpL4* using the primers mmpL4_P935_Fw and mmpL_P935_Rv. The final constructs were validated by Oxford Nanopore sequencing. A 3xFLAG tag was introduced to the pML4324 vector by Gibson assembly using primers FLAG_fw and FLAG_rv for detection of protein production levels by Western Blotting. Then the *mmpS4/mmpL4-cys-depleted* was subcloned into the pML4324_3xFLAG vector by Gibson assembly.

Cross-linked MmpS4-MmpL4 purification

The MmpS4-MmpL4 cysteine cross-linking constructs were expressed in *E. coli* MC1061. Culture volumes were either 0.6 L TB supplemented with 100 µg/ml ampicillin for small scale purification to determine crosslinking efficiency or 3 L medium to purify sufficient material for cryo-EM analyses. Cells were incubated at 37 °C for 2 h and 25 °C for 40 minutes after which protein expression was induced with 0.02% L-arabinose. Cells were harvested after 16 hours by centrifugation for 15 minutes at 4 °C and 8,000 g in a F9-6×1000 LEX centrifuge rotor (ThermoScientific) and resuspended in PBS, pH 7.2. DNase I was added and cell were lysed by three to four passages through a Microfluidizer M-110P (Microfluidics) at 25 kPa on ice. Cell debris was removed by centrifugation for 15 minutes at 4 °C and 10,000 g in a Sorvall SLA-1500 rotor (ThermoScientific). Membrane vesicles were harvested by ultracentrifugation for 2 h at 170,000 g in a Beckman Coulter ultracentrifuge using a Beckman Ti45 rotor and resuspended in 10 or 30 ml PBS, pH 7.2 supplemented with 10% glycerol. Prior to solubilization, 10 mM N-ethyl-maleimide was added and the resuspended membranes were rotated at 22 °C for 2 h. Then proteins were extracted by solubilizing the membranes using 1 % (w/v) β-DDM (Glycon Biochemicals) for 2 h at 4 °C. Unextracted material was harvested by ultracentrifugation at 170,000 g for 20 min and 0.5 ml of NHS activated Sepharose 4 Fast Flow (Cytiva) coupled to the anti-GFP nanobody (PDB-ID: 3K1K) was added to the supernatant to capture the GFP tagged MmpL4-MmpS4 complex. All purification steps were carried out at 4 °C. Batch binding was performed for 2 h, after which the resin was washed with 20 ml of PBS, 10 mM NEM, 10 % glycerol, 0.03 % β-DDM followed by 50 ml TBS (20 mM Tris/HCl, pH 7.5, 150 mM NaCl), 10% glycerol, 0.03% β-DDM. The resin was incubated with 100 µg 3C protease for 2 h to elute the protein by cleaving off the GFP-His10-tag. The flow through was collected and the resin was washed with 2-4 ml buffer to collect the cleaved protein. The sample was analyzed by reducing and non-reducing SDS-PAGE gel electrophoresis to assess crosslinking efficiency. Proteins were concentrated with 100 kDa MWCO concentrator (Merck Millipore) to 1-2 mg/ml and flash frozen and stored at -80 °C or immediately purified by size-exclusion chromatography on a Superose 6 Increase 10/300 GL column (Cytiva) in TBS, 0.03 % β-DDM.

Cryo-EM sample preparation and data acquisition

The cross-linking construct #5 of the MmpS4-MmpL4 complex was concentrated to 4 mg/ml and applied to glow discharged (0.39 mbar, 15 mA, 1 min) holey carbon grids (Quantifoil R1.2/1.3 Au 300 mesh). Excess sample was removed by blotting at 4 °C for 4.5-5.5 s with a relative humidity of 100% and a blotting force of 20 using a Vitrobot Mark IV (ThermoFisher). The sample was vitrified by plunging into ethane/propane. Cross-linked MmpS4-MmpL4 was imaged on a Titan Krios G3i (300 kV, 100 µm objective aperture) equipped with a BioQuantum energy filter (20 eV energy filter slit) and a K3 direct electron detector (6k x 4k pixels). EPU 2.9 was used for automated data

acquisition of dose-fractionated micrographs in a defocus range of -1 – -2.2 μm in 0.2 μm increments. Micrographs were recorded at a magnification of $130,000\times$, with a pixel size of 0.65 $\text{\AA}$ and 1.3 s exposure time in counted super-resolution mode and Fourier cropped by 2. A total of 14588 micrographs were recorded for the cross-linked MmpL4-S4 complex #5 with a dose of 21.10 $\text{e}^-/\text{px}/\text{sec}$ resulting in a total electron exposure of 62.92 $\text{e}^-/\text{\AA}^2$.

Cryo-EM image processing, model building and refinement

The acquired data was processed in CryoSPARC4.7.0 [43]. Data quality was monitored and initial processing steps including patch motion correction and patch CTF estimation were performed in CryoSPARC live. Good micrographs were selected based on the resolution of the estimated CTF fit and relative ice thickness. Particles were selected and filtered in two different ways (Fig. S4 [43]). First, particles were picked using the blob picker, extracted with a box size of 400 pixel and Fourier cropped to 100 pixels (2.6 $\text{\AA}$ per pixel). Two rounds of 2D-classification were performed, followed by one ab-initio reconstruction and one additional round of 2D-classification. In a parallel approach, micrographs were first denoised and then particles were picked using the blob picker. The resulting particles were selected in the same way as described above, with two rounds of 2D-classification, one round of ab-initio reconstruction and one additional round of 2D-classification. The final selection of particles from both approaches were merged and duplicates were removed. Particles were unbinned, reextracted, and an ab-initio reconstruction was performed. The subsequent non-uniform (NU) refinement [44], was performed once with C1 symmetry and once with C3 symmetry constraints. NU refinement using C1 symmetry resulted in a 3.09 $\text{\AA}$ map of the MmpL4-S4 complex with a “short” CCD. NU refinement using C3 symmetry resulted in a 2.89 $\text{\AA}$ map of the MmpL4-S4 complex. The C1-symmetrical map was locally refined with a mask created from its CCD. The local refinement resulted in a 3.22 $\text{\AA}$ map of the MmpL4-S4 complex with a “long” CCD. Resolutions are stated according to the 0.143 cut-off criterion. Local resolution estimation was carried out in CryoSPARC.

Model building

The monomeric MmpL4 model (PDB: 9GI0) was docked into the C3 symmetrical density obtained for the MmpL4-S4 complex using PHENIX [45], Coot [46] and ISOLDE [47]. This model was used as input for the C1-symmetrical model with “short” and “long” CCD and the AlphaFold3 prediction of the CCD of the MmpL4-S4 complex was docked into the tilted CCD density. Cavities of MmpL3, the monomeric MmpL4, and the MmpL4-S4 complex were calculated using a local installation of the 3 V package (shell probe radius: 5 $\text{\AA}$, solvent-excluded probe radius of 1.4 $\text{\AA}$, grid spacing 1.3) [27].

Mycobactin toxicity and bedaquiline susceptibility assays

The MmpS4-MmpL4-cys-depleted construct was analyzed in the mycobactin toxicity assay as described previously [9]. The susceptibility of the *Mtb* Δ mmpL4/mmpL5 strain or the *Mtb* Δ mmpL4/mmpL5 complemented with *mmpS4-mmpL4-cys-depleted* were measured using a microplate Alamar Blue assay (MABA). Briefly, cultures of the *Mtb* Δ mmpL4/mmpL5 strain complemented with *mmpS4-mmpL4-cys-depleted* containing a C-terminal FLAG-tag were grown in supplemented 7H9 medium to an OD600 of 1. The cultures were then filtered using a 5 μm filter to remove clumps and cells were resuspended in Hartmans de Bont (HdB) minimal medium supplemented with 0.2% casamino acids, 24 $\mu\text{g}/\text{ml}$ pantothenate, 20 μM hemin and 0.02% tyloxapol. The cells were then incubated in microplates with increasing mycobactin concentrations for five days at 37 $^{\circ}\text{C}$. Resazurin (90 μM) was added and cells incubated for an additional six hours after which the fluorescence of the metabolically converted resazurin dye was measured as an indicator of cell viability.

Supplementary information

	MmpS4-MmpL4 “C3 symmetrical” (EMD-55355, PDB: 9SYV)	MmpS4-MmpL4 “short CCD” (EMD-55350, PDB: 9SYJ)	MmpS4-MmpL4 “long CCD” (EMD-55353, PDB: 9SYT)
Data Collection & Processing			
Microscope	Titan Krios G3i		
Camera	Gatan K3 GIF		
Magnification	130,000		
Voltage (kV)	300		
Electron exposure (e ⁻ /Å ²)	62.92		
Defocus range (μm)	-1 to -2.2		
Pixel Size (Å)	0.65		
Initial number of micrographs (no.)	14,588		
Initial Particle Images (no.)	8,987,065		
Final Particle Images (no.)	76,903	76,903	76,903
Symmetry Imposed	C3	C1	C1
Map Resolution (Å)	2.89	3.09	3.22
FSC Threshold	0.143	0.143	0.143
Map Resolution Range (Å)	2.8 – 4.0	2.9 – 6.5	3.0 – 8.0
Refinement			
Initial Model Used	9GI0 and AF model	9GI0 and AF model	9GI0 and AF model
Model Resolution (Å)	2.9	3.1	3.2
FSC Threshold	0.143	0.143	0.143
Model Resolution Range (Å)	2.8 – 4.0	2.9 – 6.5	3.0 – 8.0
Map sharpening b-factor (Å ²)	-78.2	-62.6	-63.9
Model Composition			
Non-hydrogen Atoms	35,235	37,508	38,803
Protein Residues	2,274	2,422	2,507
Ligands	LMT: 3	LMT: 3	LMT: 3
B Factor (Å ²)			

Protein	50.34	66.47	153.81
Ligand	56.69	67.39	124.06
R.m.s deviations			
Bond Lengths (Å)	0.004	0.004	0.004
Bond Angles (°)	0.716	0.630	0.653
Validation			
MolProbity Score	0.94	1.01	1.16
Clashscore	1.79	2.35	2.73
Poor Rotamers (%)	0.50	0.00	0.00
Ramachandran Plot			
Favoured (%)	98.01	98.33	97.51
Allowed (%)	1.99	1.67	2.49
Disallowed (%)	0	0	0

Table S1. Cryo-electron microscopy data collection statistics

Sequence name	Sequence (5'-3')
mmpS4-mmpl4-cysteine-depleted	CTAATGCGGACTTGGATTCCACTGGTCATCCTGGTGGTGGTCATCGTCGGGGGCTTACCCTGTCACCGGATCCGCGGCTTCTTCGGCTCCGAAAACCGCCGTCGTACTCCGACACCAACCTGGAAAACAGCAAACCATTC AACCTAAACACCTGACTTACGAGATCTTCGGACCCCGGAACAGTCGCGGACATTAGTTATTTTCGACGTCAATTCGAGCCGCAACGGGTCGATGGAGCGGTGCTACCGTGGTCATTGCATATCACGACAAACGACGCGCGGTGATGGGAAATATCGTGGCACAAGGTAATAGCGACAGCATTGGCTGCCGAATCACGGTAGACGGCAAGGTCAGGGCCGAGAGGGTTTCCAACGAAGTCAACGCCTATACTTACTGCTTGGTGAAGTCCGCGTGAGTACTAAATTCGCGAACGACTCCAATACCAACGCTCGCCCCGAAAAGCCATTATCGCCAGGATGATCCACGCCCTTTGCGGTACCGATCATCCTTGGCTGGCTGGCAGTTTCCGTCGTGTGCACCGTATTTGTCCTGCTACTGGAAGCTGTCGGCCAAGAGCGATCGGTGTCGCTGAGTCCCAAGGATGCACCGTCGTTTGAGGCGATGGGACGTATCGGCATGGTGTTCAAGGAAGGCGATTCCGACAGTTTCGCGATGGTCATAATAGAGGGTAACCAACCCCTTGGCGACGCTGCCATAAGTATTACGACGGCCTGGTTGCTCAATTGAGGGCCGATAAGAAGCACGTGCAAAGTGTCCAAGATTTATGGGGGGACCCACTCACCGCCGCGGGCGTGCAAAGTAACGACGGCAAGGCCGCCTATGTTCACTGTCACTTGGCGCAACCAAGGCACGCCGCTGGCCAACGAATCCGTCGAGGCAGTACGCAGCATCGTCGAAAGCACGCCCGCGCCGGGCATAAAGGCCTATGTGACCGGACCATCCGCACTCGCCGCGGATATGCCACAGTGGCGATAGATCCATGGCCAGGATCACCATGGTTACGGTCCGCGTGATCTTTATTATGTTGTCGCTCTACCGTTCGATAATCACCGTGGTTCTCTGCTGATCACGGTGGGGGTGCAATTGACGGCTGCGCGCGGAGTCTGATCGGTTCTGGGGCATAGCGGGGCTATCGGACTTACCACCTTTGCGGTGAGCTTGCTGACTTCGCTGGCAATCGCGGCCGTACGGACTACGGGATATTATCATCGGGCGCTACCAGGAAGCCGCCAAGCCGGCGAGGACAAAGAGGCCGCCTACTACACCATGTACCGGGGACCGCCACGTGATTCTGGGCTCTGGATTGACCATCGCCGAGCTACCTTTCCCTGAGCTTTGCCCGCATGCCCTACTTTCAAACCTGGGCATCCCTCCGCGGTGGGGATGCTAGTCGCAGTGGCGTGC GTTGACGCTGGGTCCGGCCGTCTACACGTCGGCAGCCGGTTCGGCTGTTTCGACCTAAGCGGCTCCTCAAAGTCCGTGGCTGGCGCGGGTGGGTACCGTGGTGGTTCGCTGGCCGTGCCGTTCTCGTCGCCACCTCCGCGATCGCCCTAGTCGGTCTGCTCGCCCTGCCTGGATACAAAACAGCTACAACGACCGTGACTACCTACCGGACTTCATACCGCCAACCAAGGATATGCGGCCCGGATCGTCATTTCTCTCAGGCCAGGATGAAGCCGAGATCCTAATGATTGAATCCGATCACGATATGCGGAATCCGGCAGATTTCTGGTCTGGACAAGCTGGCTAAAGGCATCTCCGGTCCCGGGCATTTCCGTGTGCAAGCAATTACCGGCCGAAGGAACGACGATGGACCAACGTCGATCCCGTTCAGATCAGCATGCAA AATGCCGGTCAGCTGCAGACCATAAAGTACCAGCGCAGCGGGCGAACGACATGCTGAAGCAGGCCACGAGATGGCCAGCAGATTGCGGTGTTGACGCGGATGCATAGCTTGATGGCGGAGATGGCCAGTACTACCACCGCATGGTCG

Table S2. Nucleotide sequence of *mmpS4-mmpl4_cys-depleted*

GCGACACCGAAGAGATGAAGGAGATCACCGAAGAATTACGCGACCACTGCGGGATTTTGATGATTTCTGGAGACCA
AATTCGAAGCTATTTCTACTGGGAAAAGCACTGCTACGGAATTCGATCTGTTGGTCTGTTGAGATCGATATTCGATGC
ACTGGACGGAATCGACAAGCTCAGCGAGCAATAGGCGTCTCTTGGGCGACTTGGCGGAGATGGATCGCCTCATG
CCACAATGGTCGCGCAGATCCCGCCGAGATTGAGGCCATGGAAAACATGCGGACCATGATCTTACCATGACACA
GCACCATGACCGGAATCTTCAGCAGATGCTCGAAATGAGCGACAATGCCACAGCCATGGGTAAAGCTTTTGACGC
CGCCAAGAACGACGACTGTTCTACCTTCCACCCGAGGTTTTCAAGAACAAAGACTTCCAGCGCGCCATGAAGTCGT
TCCTGTCTTCGACGCGACGCGGCCGCTTTATCATCTGACAGGGGGATCCGCAATCACCCGAGGGCATCAA
AAGTATCGACGCGATTCTGACCGCGCCGAGGAATCGCTCAAGGGAATCCACTAGAAAGACGCCAAGATCTACCTG
GCCGCGACGCGCGCGTCTTCCACGACATCTCCGAGGGCGCCCAATGGGACCTTCTATCGCGGCAATTCGTCCCT
CTCTCTCATTTTATCATCATGCTGATCATCACGGGCGCTTATTGCCGCGCGCTCATTGTGGGCGACAGTCGCACCT
TCACTGGTGCTCTTTCCGACTATCGGTGCTGCTCTGGCAGCATATTCTCGCAATCCATCTGCAATGGCTCGTCT
GCGATGTCGGTCATGTTCTGTTGGCGGTGGGATCTGACTACAATCTGCTCTTGGTCTCCCGTTCAAACAGGAAAT
AGGAGCGGGGCTGAAGACCGGCATCATCCGGTCGATGGCGGACCGGCAAGGTCGTGACCAACGACGGGCTGG
TATTCGCCGTACCATGGCTCCATGGCGTCAGCGATCTGCGGGTATCGGACAGGTGGGACCAACCATCGGTCTG
GGTTTGTCTTCGACACGTTGATCGTGCATGTTCTATGACGCCATCCATCGCCGCACTGCTGGGCGTTGGTTTTG
GTGGCCGCTGCGGTTTCGACGCGTCCAGCCGTACCCGACCGTCCGAGTGAAACGCAACCGCGGGTCTGTCG
TTAGCGATGAGCAGCGATGCTCGGGTGAAGTCTAATGCGGACTTGGATTCCATGCTCATCTGGTGGTGGTCACTG
TCGGGGGCTTACCCTGACCGGATCCGCGGCTTCTCGGCTCCGAAAACCGCCGCTGCTACTCCGACCAACCTG
GAAAACAGCAACCAATCAACCTAAACACCTGACTTACGAGATCTTCGACCCCCGGAACAGTCGCCGACATTAG
TATTTTCGACGTAATCCGAGCGCAACGGGTGATGGAGCGGTGCTACCGTGGTCTTGCATATCACGACAAACG
ACGCGCGGTGATGGGAAATATCGTGCAAGTAATAGCGACAGCAATGGCTGCCGAATCACGGTAGCGGCA
AGGTGAGGGCGAGAGGGTTTCAACGAAGTCAACGCTATACTTACTGTTGGTGAAGTCCGCGTGAGTACTAAA
TTCGCAACGACTCAATACCAACGCTCGCCCGAAAAGCCATTATCGCCAGGATGATCCAGCCTTTGCGGTACC
GATCATCTTGGTGGTGGGAGTTTCCGTGTTGTACCGTATTTGTCGCGTCACTGGAAGCTGTGCGCAAGAGC
GATCGGTGCTGCTGAGTCCAAAGGATGACCGTCTTTGAGGCGATGGGACGATCGGCGATGTTGTTCAAGGAAG
GCGATTCCGACAGTTTCGCGATGGTCATAATAGAGGGTAACCAACCCCTTGGCGACGCTGCCATAAGTATTACGAC
GGCCTGTTGCTCAATTGAGGGCCGATAAAGACGCTGCAAGATGTTCAAGATTATGGGGGAGCCACTCACCG
CGCGGGGTGCAAAAGTAACGACGCGCAAGGCGCCTATGTTCAACTGTCACTTCCGCGCAACCAAGGACGCGCT
GGCCAACGAATCCGTCGAGGCGATGACGAGCATGTCGAAAGCACGCGCGCGCGGCGATAAAGGCCTATGT
GACCGGACCATCCGCACTCGCCGCGGATGACCAACAGTGGCGATAGATCCATGGCCAGGATCACCATGGTTACG
GTGCGCGTGATCTTTATATGTTGTTGCTGCTACCGGTGATAATACCGTGGTCTCTGCTGATCACGGTGGGG
GTCGAATTGACGCTGCGCGCGGAGTCGAGCGGTTCTGGGGCATAGCGGGGCTATCGGACTTACCACTTTGCGG
TGAGCTTGCTGACTTCGCTGCAATCGCGCGGTACGGAATACGGGATATTATCATCGGGCGCTACCGGAAGC
CCGCAAGCCGCGGAGGACAAAGAGGCGCCTACTACCATGATACCGCGGAGCCGCCACGTGATTCTGGGCTCT
GGATTGACCATCGCCGAGCTACCTTTCCCTGAGCTTTGCCGCGATGCCCTACTTTCAAACCTGGGCGATCCCTCC
GCGGTGGGGATGCTAGTCGAGTGGCGGTGCGGTTGACGCTGGGTCCGCGCGTCTACACGTGCGGACGCGGTTT
GGCCTGTTGACCTAAGCGGCTCTCAAAGTCCGTGGTGGCGCGGGTGGGTACCGTGGTGGTCTGCTGGCCG
TGCCCGTCTGCTGCGCACCTCCGCGATCGCCCTAGTCGGTCTGCTGCCCTGCTGGATACAAACACGCTACAACG
ACCGTGACTACCTACCGGACTTCAACCCGCCAAGGATATGCGGCGCGGATCGTCATTTCTCAGGCCAGG
ATGAAGCCGAGATCCTAATGATTGAATCCGATCACGATATGCGGAATCCGGCAGATTTTCTGCTCTGGAACAGT
GGCTAAAGGATCTTCCGGGTCCCGGGCATTTCCCGTGTGCAAGCAATACCGGCGCGAAGGAACGACGATGGAC
CACAGTCGATCCCGTTCAGATCAGCATGCAAAATGCGGTGAGCTGACAGCATAAAGTACAGCGCGACCGGG
CGAACGACATGCTGAAGCAGGCCGACGAGATGGCCACGACGATTGCGGTGTTGACGCGGATGCATAGCTTGATGG
CGGAGATGGCCAGTACTACCCACCGCATGTCGCGCACACCGAAGAGATGAAGGAGATCACCGAAGAATTACGCG
ACCACGTCGCGGATTTGATGATTTCTGGAGACCAATCGAAGCTATTTCTACTGGGAAAAGCACTGCTACGGAATT
CCGATCTGTTGGTCTGTTGATGATTCGATGCACTGGACGGAATCGACAAGCTCAGCGAGCAAAATAGGCGTCT
CTTGGCGACTTGGCGGAGATGGATCGCTCATGCCAAATGGTCGCGCAGATCCCGCCGAGATTGAGGCCATG
GAAAACATGCGGACCATGATTCTTACCATGCACAGACCATGACCGGAATCTTCAGACAGATGCTCGAAATGAGCG
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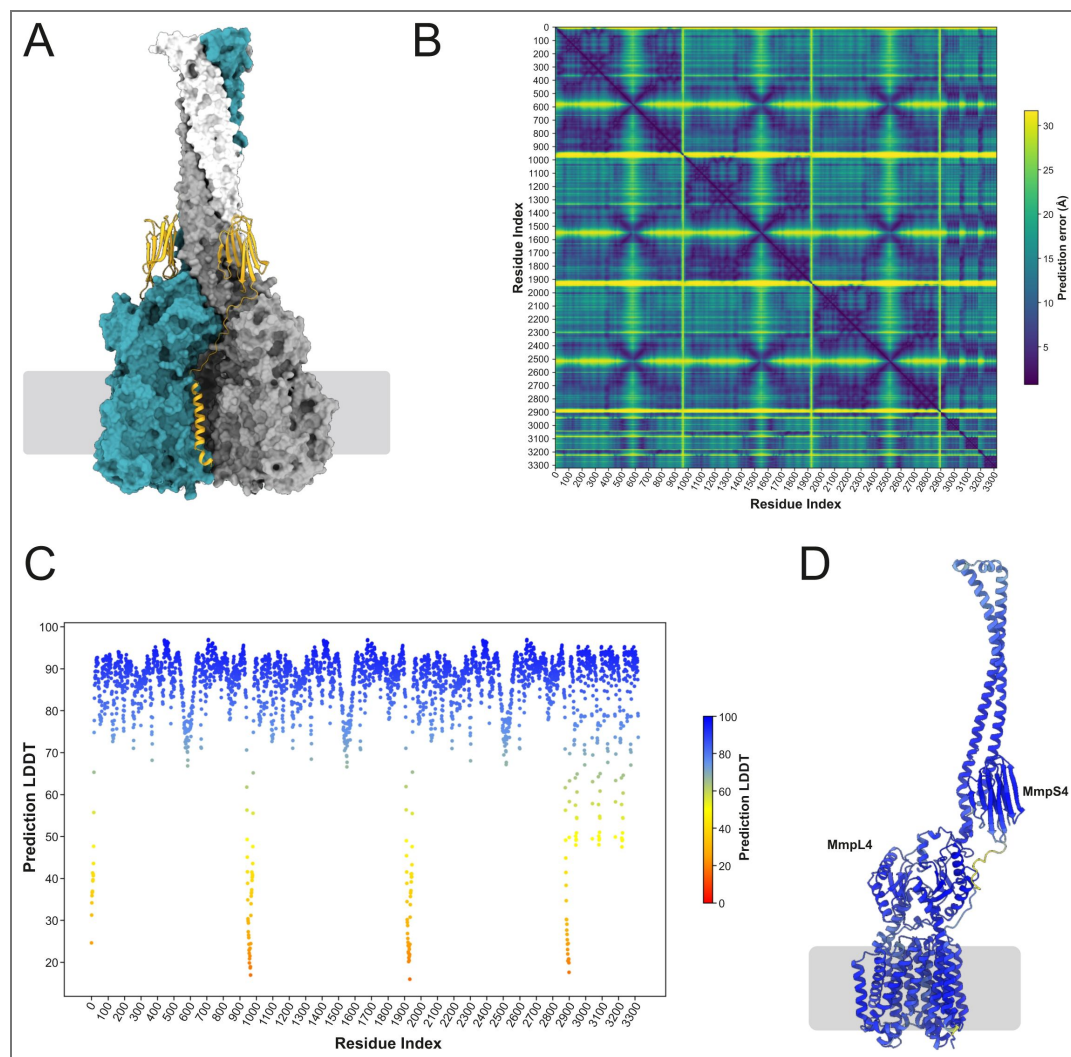
Table S2. (continued)

Primer name	Sequence (5'-3')
mmpS4_W6_Fw	AATGCGGACTTGTATTCCACTGGTCATCC
mmpS4_W6_Rv	CCAGTGGAATACAAGTCCGCATTAGACTC
mmpS4_F29_Fw	CGCGGCTTCTGCGGCTCCGAAAACC
mmpS4_F29_Rv	CGGAGCCGCAGAAGCCGCGGATCC
mmpS4_Y37_Fw	GCCCGTCGTGCTCCGACACCAACCTG
mmpS4_Y37_Rv	GTGTCGGAGCACGACGGGCGTTTTCG
mmpS4_D39_Fw	GTCGTACTCCTGCACCAACCTGGAAAACAGC
mmpS4_D39_Rv	CAGGTTGGTGCAGGAGTACGACGGGC
mmpS4_N44_Fw	CAACCTGGAATGCAGCAAACCATTC AACCC
mmpS4_N44_Rv	ATGTTTTGCTGCATTCCAGGTTGGTGTCTGG
mmpS4_A95_Fw	GACAAACGACTGCGCGGTGATGGGAAATATC
mmpS4_A95_Rv	CATCACCGCGCAGTCGTTTGTCTGATATG
mmpS4_S139_Fw	GGTGAAGTGCGCGTGAGTACTAAATTCGC
mmpS4_S139_Rv	GTA CTACGCGCACTTCACCAAGCAGTAAG
mmpL4_P935_Fw	TTGGTGGTGTCTGCGGTTCTGCAGC
mmpL4_P935_Rv	CGCGCAGACACCACCAAAACCAACGGC
mmpL4_R324_Fw	GCTTTGCCTGCATGCCCTACTTTCAAACC
mmpL4_R324_Rv	GTAGGGCATGCAGGCAAAGCTCAGGGAAAAG
mmpL4_D64_Fw	GTCCCAAGTGTGCACCGTCGTTTGAGG
mmpL4_D64_Rv	GACGGTGCACACTTGGGACTCAGCG
mmpL4_K742_Fw	GAATCGCTCTGTGGAACCTCACTAGAAG
mmpL4_K742_Rv	GGAGTTCCACAGAGCGATTCTCTCGG
mmpL4_S434_Fw	GTCATTTCTGTCAGGCCAGGATGAAGCCC
mmpL4_S434_Rv	CCTGGCCTGACAGAAATGACGATCCGC

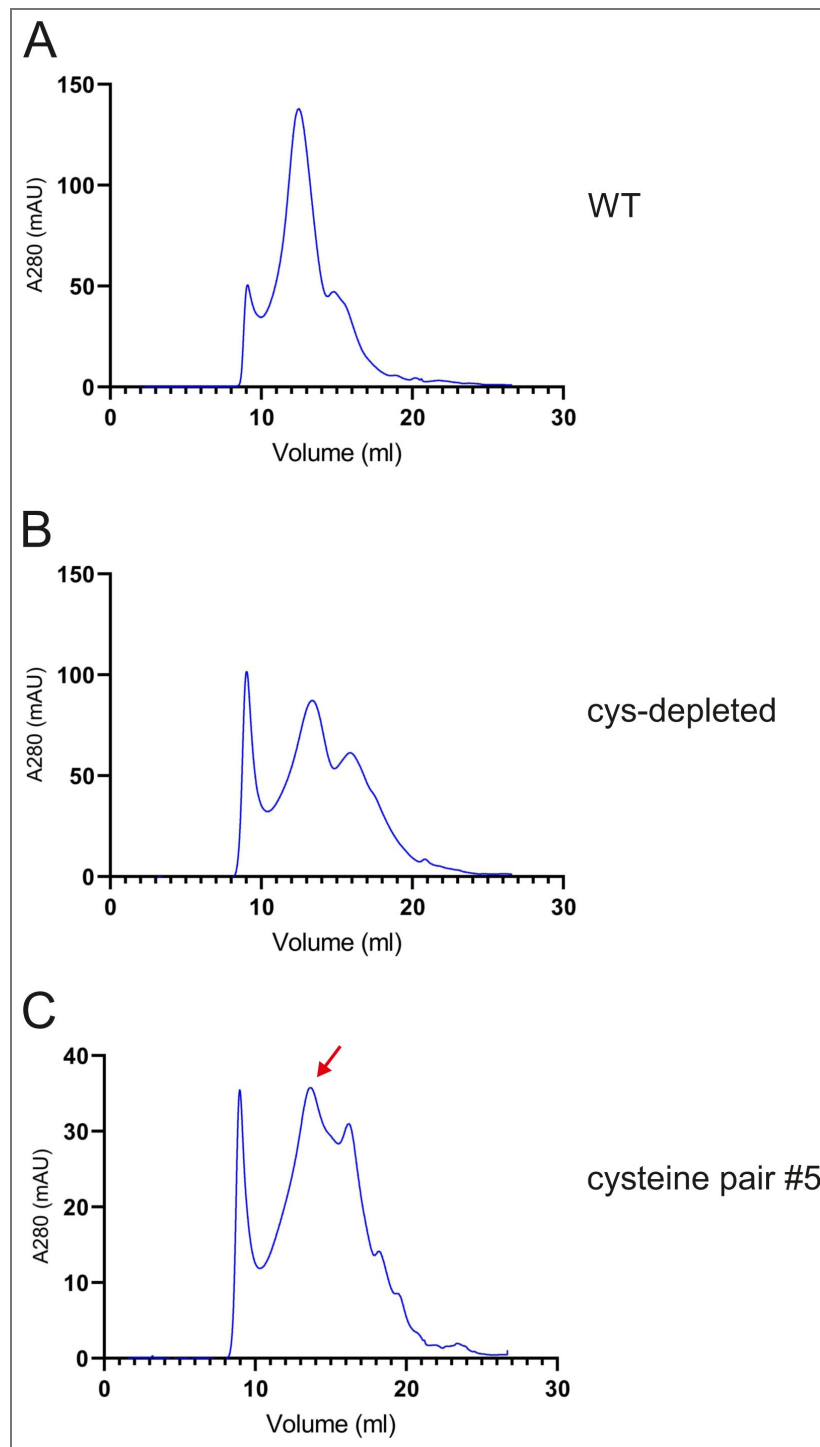
Table S3. Primers used in this study.

mmpL4_E483_Fw	CCGGCCCTGTGGAACGACGATGGACC
mmpL4_E483_Rv	CGTCGTTCCACAGGGCCGGTAATTGC
mmpL4_M487_Fw	GGAACGACGTGTGACCACACGTCGATCC
mmpL4_M487_Rv	CGTGTGGTCACACGTCGTTCTTCGGG
mmpL4_V690_Fw	CCACCCGAGTGTTC AAGAACAAGACTTC
mmpL4_V690_Rv	GTTCTTGAAACACTCGGGTGAAGGTAG
FLAG_Fw	AAA GAT CAT GAT ATC GAT TAC AAG GAT GAC GAT GAC AAG TGA GAA GCT TTA GCT AAT TAA TTG GGG
FLAG_Rv	ATATCATGATCTTTATAATCACCGTCATGGTCTTTGTAGTCgccgaggcggtcgctg
pML4324_backbone_Fw	GCA GCG ATC GTC TGG GTG ACT ACA AAG ACC ATG ACG G
pML4324_backbone_Rv	GAA TCC AAG TCC GCA TTA GCA CTA TTA ACC TCC TTT CTG TTA ATT AAG C
mmpS4L4_Fw	TAA CAG AAA GGA GGT TAA TAG TGC TAA TGC GGA CTT GGA TTC
mmpS4L4_Rv	CAT GGT CTT TGT AGT CAC CCA GAC GAT CGC TGC

Table S3. (continued)

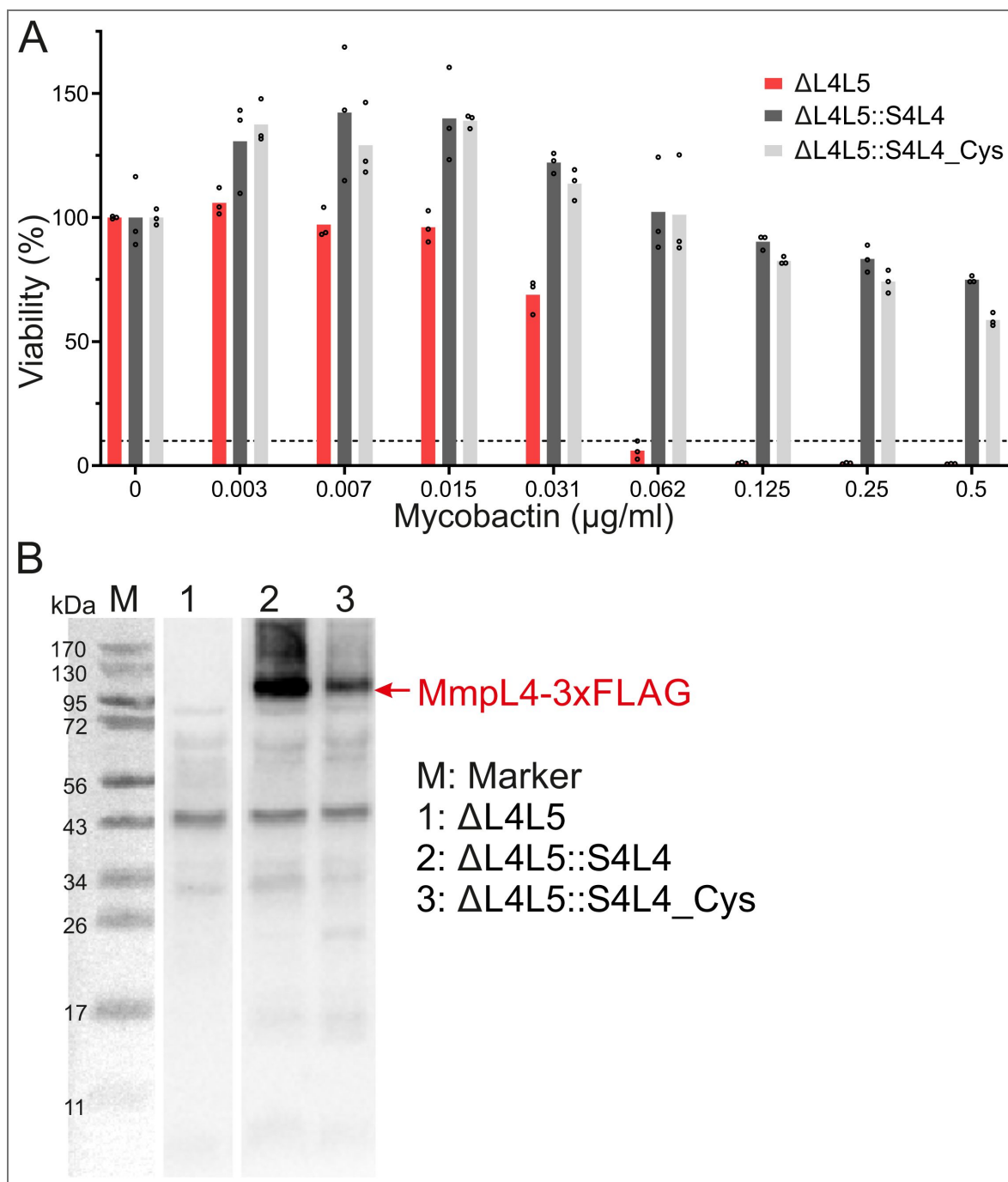


Supplementary Figure 1. AlphaFold2 prediction of the hexameric (MmpS4)₃-(MmpL4)₃ complex. (A) The three protomers of MmpL4 are shown in surface presentation and colored petrol, grey and white. The MmpS4 protomers are shown as yellow cartoon (only two of them are visible). The boundary of the cytoplasmic membrane is indicated by grey lines. (B) Predicted alignment error (PAE) plot of the hexameric (MmpS4)₃-(MmpL4)₃ complex reported by AlphaFold2. It shows the expected distance error (in Å) between each residue in the predicted model. The three MmpL4 protomers cover residues 1-2901, and the three MmpS4 protomers residues 2902-3321. (C) Predicted Local Distance Difference Test (pLDDT) plot. The three MmpL4 protomers cover residues 1-2901, and the three MmpS4 protomers residues 2902-3321. (D) Depiction of pLDDT values in the context of the predicted MmpL4/MmpS4 structure (only first protomer shown, N- and C-terminus of MmpL4 cleaved).



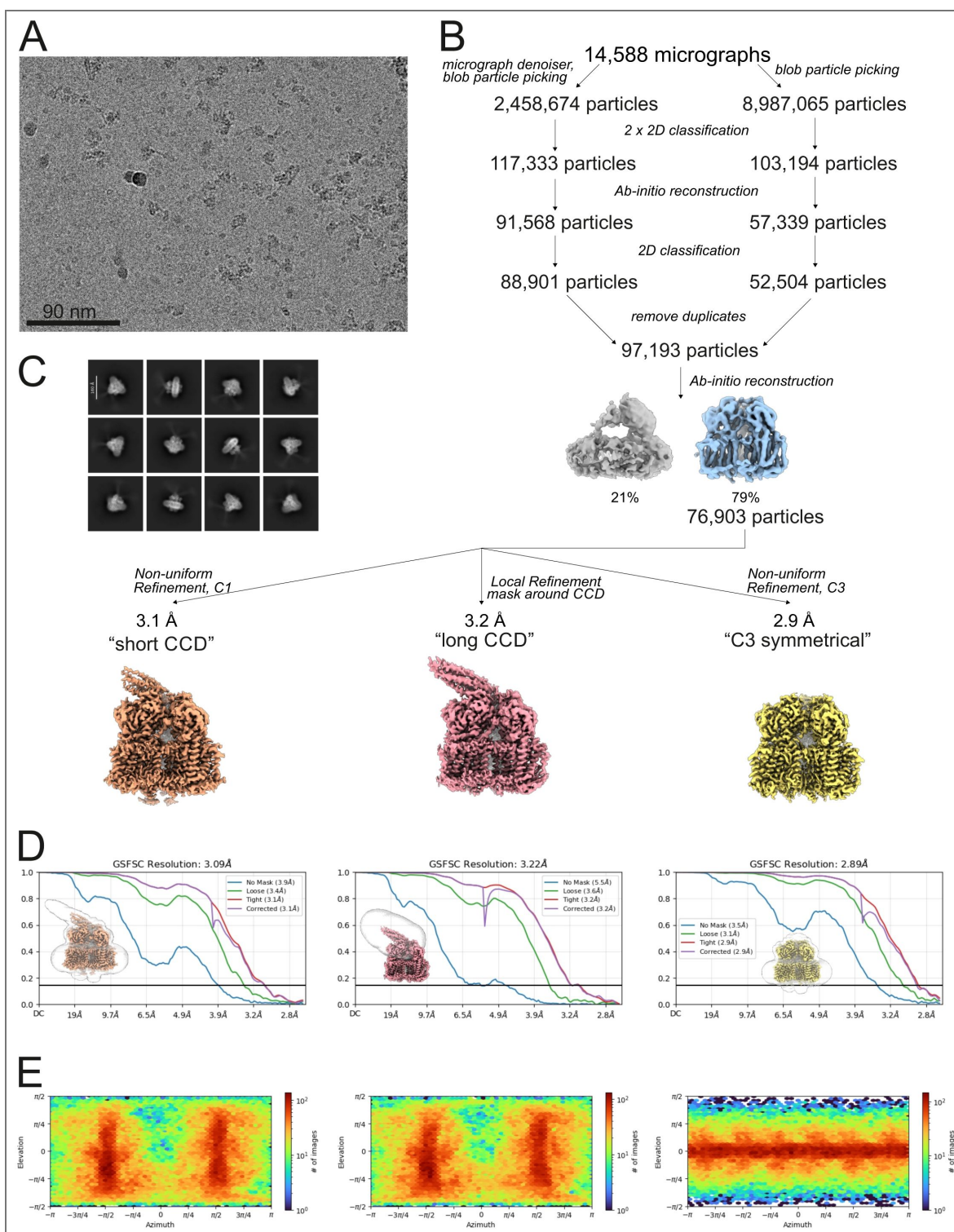
Supplementary Figure 2. Size exclusion chromatograms.

(A–C) Size exclusion chromatography traces of wild-type MmpS4-MmpL4 (A), cys-depleted MmpS4-MmpL4 (B) and MmpS4-MmpL4 containing the cysteine pair #5 (C). The proteins were separated on a Superose 6 Increase 10/300 GL column. The main peak of MmpS4-MmpL4 containing the cysteine pair #5 eluting at 13.6 ml (indicated by red arrow) was concentrated and analyzed by cryo-EM. The cryo-EM structure of the hexameric (MmpS4)₃-(MmpL4)₃ complex was determined from this sample.



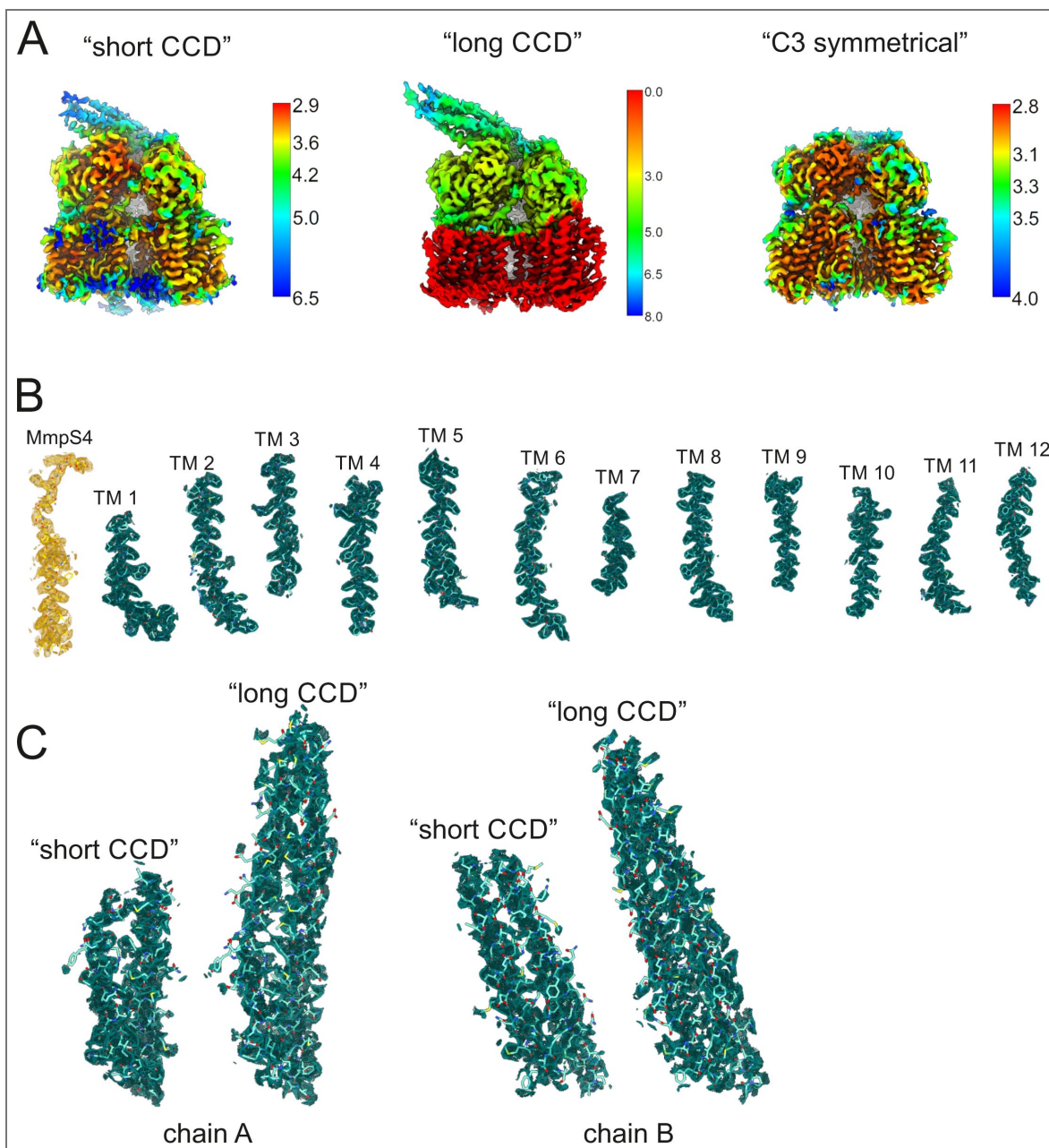
Supplementary Figure 3. Functional analysis of cysteine depleted mutant.

(A) Viability of *M. tuberculosis* strains at increasing mycobactin concentrations was determined using the microplate Alamar Blue assay (see methods). Data were normalized to the viability in the absence of mycobactin (100 %). ΔL4L5, *M. tuberculosis* strain lacking *mmpL4* and *mmpL5*; ::S4L4 complementation with *mmpS4-mmpL4*; ::S4L4_Cys complementation with cysteine depleted *mmpS4-mmpL4*. Data points correspond to technical triplicates. (B) Western blot analysis of MmpL4 extracted from the strains shown in (A) using an anti-3xFLAG antibody (see methods).



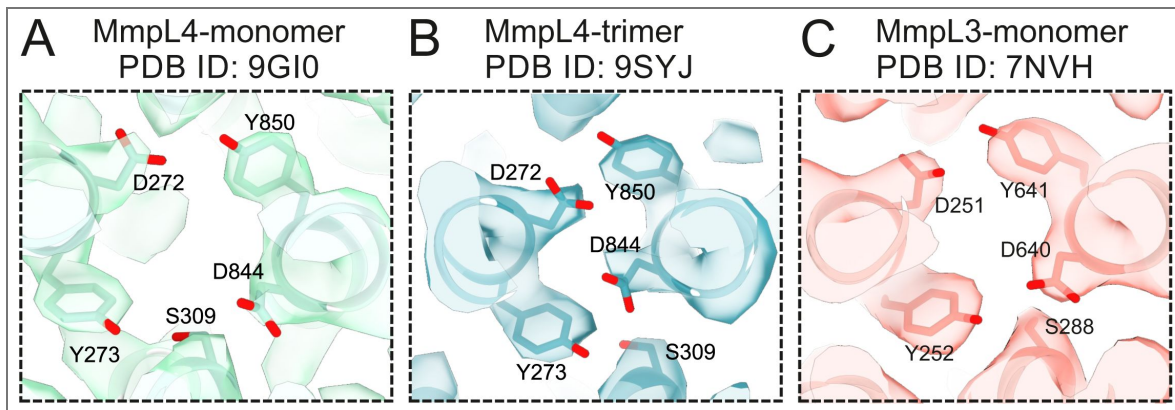
Supplementary Figure 4. Cryo-EM reconstruction of the hexameric MmpS4-MmpL4 complex.

(A) Representative cryo-EM micrograph. (B) Image processing work flow. (C) 2D-class averages. (D) FSC plot used for resolution estimation. (E) Angular distribution plot of particles included in the final 3D reconstruction.



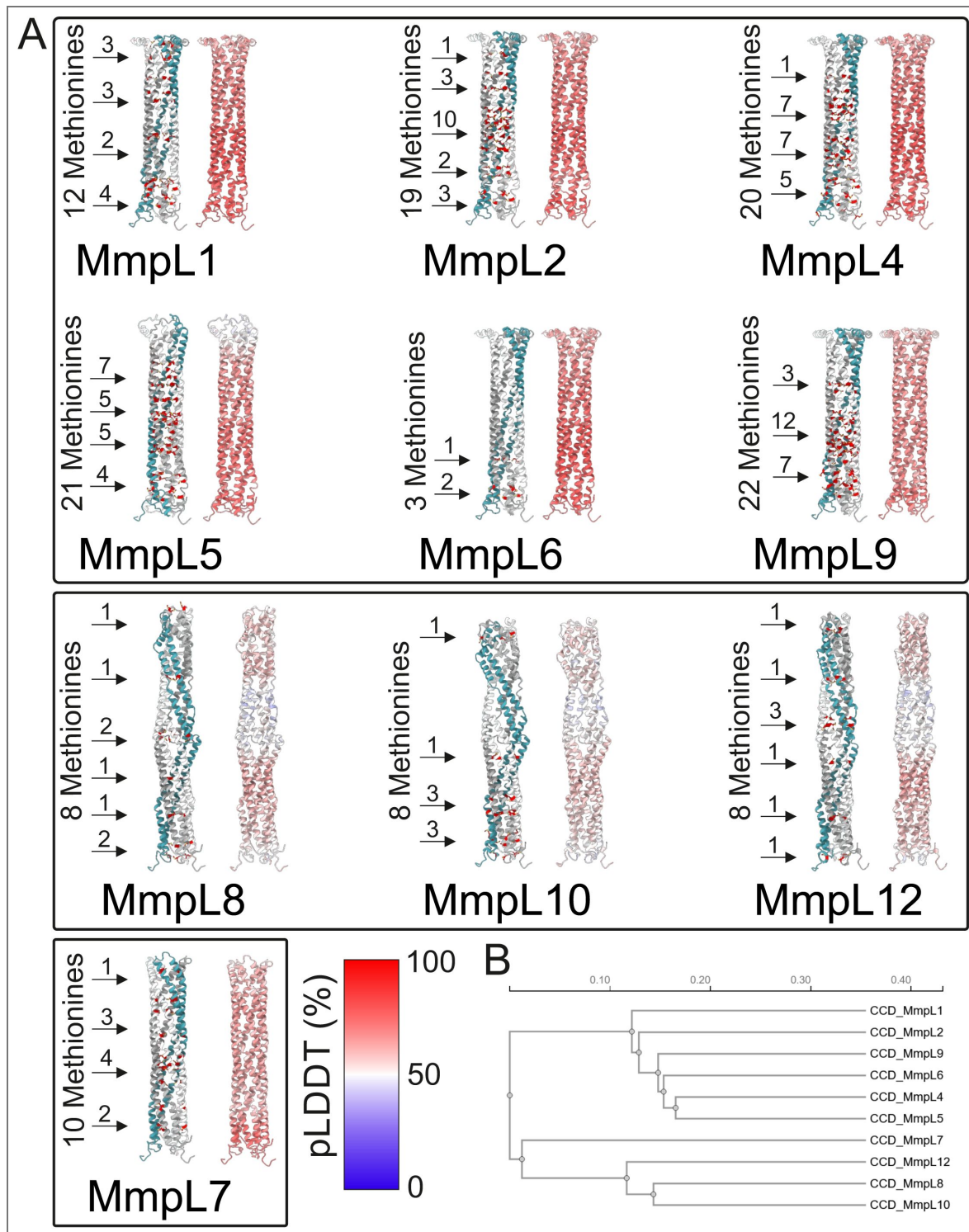
Supplementary Figure 5. Local resolution estimations and cryo-EM densities of the hexameric MmpS4-MmpL4 complex.

(**A**) Final reconstructed maps colored by local resolution. Note that for the “long CCD” map, a large part of the map was not part of the mask and thus no local resolution could be determined (red parts, resolution arbitrarily set to zero). (**B**) Cryo-EM densities of the hexameric MmpS4-MmpL4 complex “short CCD” with the respective refined model superimposed. The model is shown as sticks and structural elements are labelled. Transmembrane helices (TM) are colored in green, MmpS4 in yellow. Densities were contoured at 5.4σ . (**C**) Cryo-EM densities of CCDs from “short CCD” and “long CCD” models, chains A on the left and chains B on the right. Densities were contoured at 4.2σ . Cryo-EM reconstruction of the hexameric MmpS4-MmpL4 complex.



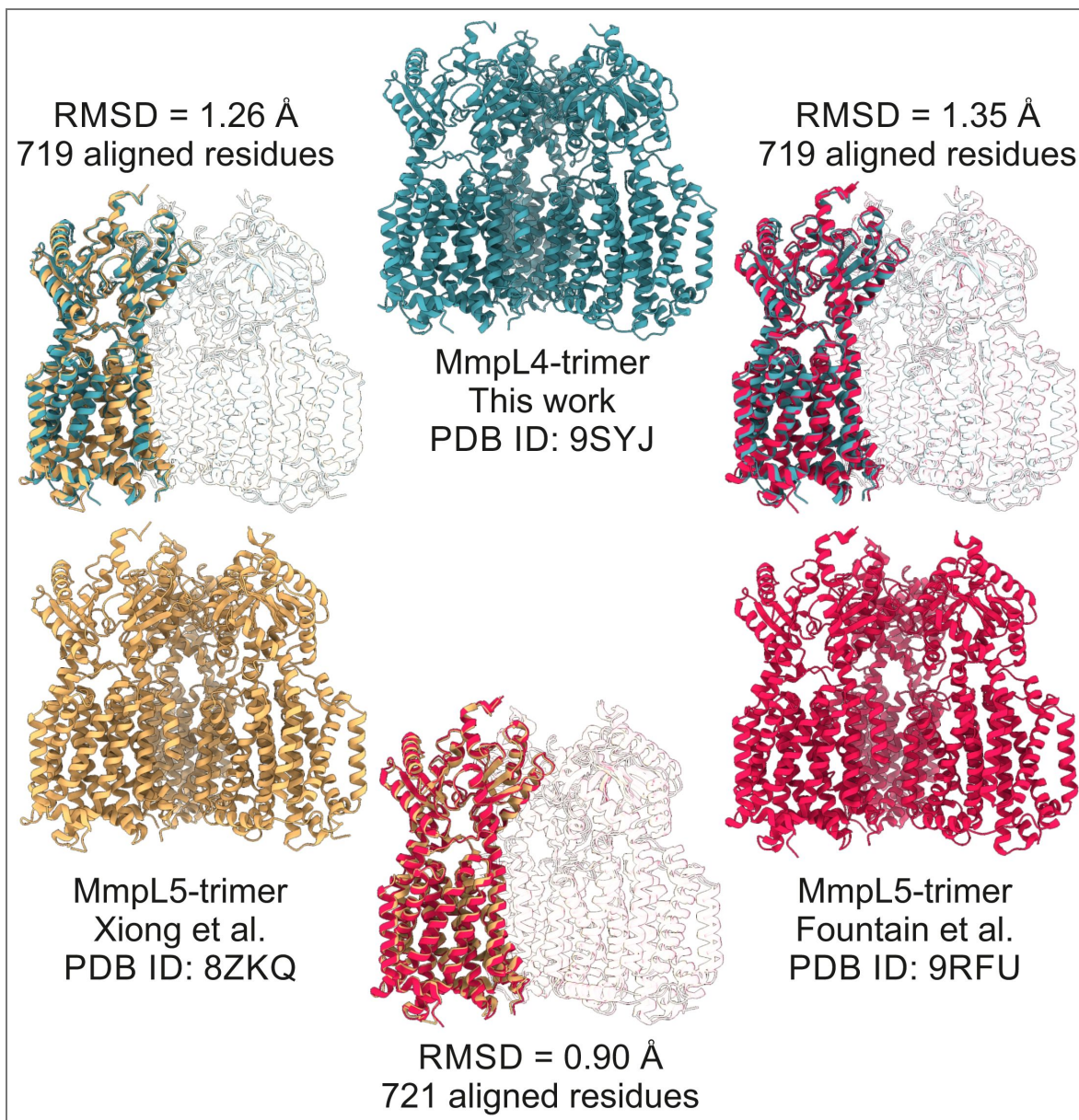
Supplementary Figure 6. Cryo-EM densities of DY-pairs.

The conserved DY-pairs and S309(MmpL4)/S288(MmpL3) are shown as sticks and viewed from the periplasm as in main Fig. 3. (A) Monomeric MmpL4 (PDB ID: 9GI0) contoured at 11.9σ . (B) Trimeric MmpL4 (this work, PDB ID: 9SYJ, chain A) contoured at 8.9σ . (C) Monomeric MmpL3 (PDB ID: 7NVH) contoured at 4.5σ .



Supplementary Figure 7. Comparative analysis of CCDs.

(A) CCDs of *M. tuberculosis* MmpL proteins shown as cartoons. The models are AlphaFold3 predictions of the respective MmpS-MmpL hexamers (in case an MmpS protein is present) or the MmpL trimers. For each CCD, methionine residues are shown as red sticks and the number of methionine residues per α -helical hairpin is provided in the pictures on the left and in the pictures on the right, the models are colored according to their Predicted Local Distance Difference Test (pLDDT) values. The coloring scheme is provided at the bottom. (B) Phylogenetic tree of the CCDs shown in (A).



Supplementary Figure 8. Structural comparisons of trimeric *M. tuberculosis* MmpL4 and MmpL5.

The first protomers of the trimeric MmpL4 structure (this study, PDB ID: 9SYJ, chain A) and the two trimeric MmpL5 structures (Xiong et al., PDB ID: 8ZKQ, chain A / Fountain et al., PDB ID: 9RFU, chain D) were superimposed using the matchmaker tool of ChimeraX (Needleman-Wunsch alignment algorithm, BLOSUM-62 similarity matrix). The respective RMSD values are depicted directly in the figure, along with the number of aligned residues.

Data availability

Structural models have been deposited to the Protein Data Bank (PDB) with the accession codes 9SYJ (data processed using C1 symmetry), 9SYT (data processed using C1 symmetry and applying a mask to improve the resolution of the CCD), and 9SYV (data processed using C3 symmetry). The cryo-EM data has been deposited in the Electron Microscopy Data Bank (EMDB) with the accession codes D_1292151303, D_1292151414, and D_1292151419 respectively. Plasmids and other data that support the findings of this study are available from the corresponding authors upon request.

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Additional information

Author contributions

J.C.E., M.N., and M.A.S. conceived the project. J.C.E. generated the cysteine variants, performed the cross-linking analysis, purified the cross-linked complex #5, and prepared cryo-EM grids. N.P.L. measured cryo-EM data, processed the data with help of A.A.G., and built the model with support of A.A.G.. V.M. performed mycobactin toxicity assays and Western blot analyses. J.C.E., N.P.L., M.N., and M.A.S. interpreted the data. N.P.L. prepared the figures with inputs from J.C.E., and M.A.S.. N.P.L., J.C.E., M.N., and M.A.S. wrote the manuscript with input from all other authors.

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Peer reviews

Reviewer #1 (Public review):

This manuscript adds to the recent, exciting developments in our understanding of the MmpL/S transporters from mycobacteria. This work provides solid support for the trimeric/hexameric arrangement of subunits in the complex, and reveals a possible pathway for substrate translocation.

Overall, I think this manuscript is a solid body of work that adds to several recent studies from this team and others on the structure and mechanism of the MmpL/S transporter family, particularly MmpL4/S4. The combination of AF, disulfide engineering, and experimental structure is good, though it is a bit puzzling that the experimental structure based on disulfide stabilization of the AF prediction does not recapitulate key elements (MmpS periplasmic domain docking to MmpL, and altered CCD configuration).

I have no major concerns about this manuscript; I could make only a few minor suggestions for clarification in a revision.

<https://doi.org/10.7554/eLife.110432.1.sa3>

Reviewer #2 (Public review):

Summary:

The manuscript describes the structure of the Mycobacterium tuberculosis (MmpS4)3-(MmpL4)3 hetero-hexameric transporter complex. The structure was obtained by cryogenic electron microscopy using an engineered construct that cross-links MmpS4 to MmpL4 via a disulfide bond. The position of the disulfide bond was determined using an AlphaFold2 model of the hetero-hexamer. Although AlphaFold2 predicts a symmetric hetero-hexamer, the author found that the structure of the coiled-coil domain (CCD) is asymmetric, tilted at about 60° relative to the membrane domains, and only contains two of the three alpha helical hairpins, with the third being disordered.

Strengths:

The strategy of using AlphaFold2 models to guide construct design for experimental structure determination is state-of-the-art, and this work provides a great example of its applications

and limitations. I.e., the experimental structure does not fully recapitulate the prediction but provides unexpected results.

The comparisons between the authors' structures and the previously published structures of the MmpL4 monomer and MmpL5 trimers strengthen the authors' findings.

Weaknesses:

A more detailed description of the current mechanistic hypothesis would strengthen the manuscript. The authors state that the two periplasmic domains "are expected to undergo rigid body movements that allow substrate transport through these periplasmic domains similar to the conformational changes observed in the *E. coli* multidrug efflux pump AcrB". A schematic of the proposed transport cycle, as a supplemental figure that shows the current hypothesis regarding transport, would be beneficial for understanding the previous structures and putting the current structure in context. Outside of "the mechanistic basis of how these conformational changes are coupled to protonation of the DY-pairs", what are the major controversies/open questions regarding the mechanism?

The authors provide evidence that the cysteine-depleted S4L4 construct is functional, but do not show that the construct with the introduced disulfide bond #5 (D39C MmpS4 and S434C MmpL4) is also functional. Demonstrating this would allow the authors to better interpret their resulting structures.

The analysis presented in Figure 5 and Supplementary Figure 7 seems to suggest that the authors are proposing that the CCD central cavity acts as a transport pathway for the transported substrate, but I am not sure that this hypothesis is explicitly stated. This makes the reasoning behind the analysis presented unclear. Clarity could be improved by stating that the hypothesis of direct transport of substrate through the CCD central channel is being examined using the structure prediction, and what the implications are for the structure solved with the incompletely formed CCD.

Given that the results emphasize the flexibility of the CCD, the manuscript would be strengthened by 3D variability analysis either in cryoSPARC or using cryoDRGN (or both). This would allow the authors to better quantify the degree of motion in the CCD and how it may correlate to flexibility in other regions. Further 3D flex reconstruction in cryoSPARC may improve the map quality of the CCD.

<https://doi.org/10.7554/eLife.110432.1.sa2>

Reviewer #3 (Public review):

Summary:

This manuscript by Earp et al reports cryoEM structures of the hexameric (MmpS4)₃-(MmpL4)₃ complex from *Mycobacterium tuberculosis*, which belongs to the RND family of transporters and is known to have a role in the export of siderophores and contribute to drug resistance. The experimental workflow showcased involves the design of disulfide pairs using distance constraints obtained from the AlphaFold predicted structure of the hexameric complex. One such disulfide pair was used to determine the ~3.0 Å structures. The structure reveals density for the previously unresolved coiled-coil domain (CCD), a tilted CCD arrangement, and a cavity within the periplasmic domain, which the authors assert is occupied by detergent. Comparison of this complex with the monomer structure of MmpL4 shows conformational variations interpreted to implicate different domains and conserved residues involved in proton coupling, which might be related to the transport mechanism. While the methodological aspects of the manuscript are solid, enthusiasm for the overall advance/significance is less so, with doubts about the relevance of the tilted CCD structure,

considering disulfide trapping and an incomplete validation of the claim that the titled CCD represents a stable intermediate conformation. A clear, updated transport mechanism is largely missing from the manuscript.

Strengths:

Beautiful structures, AF prediction-experimental validation nexus that could be fine-tuned for different systems/difficult to target complexes.

Weaknesses:

Physiological relevance of the tilted CCD conformation. No clear mechanistic model for the transport. While the CCD may indeed be a stable intermediate, the fact that the rest of the trimeric arrangement is unaffected does not fully rule out disulfide trapping as a factor in promoting this. The findings would be strengthened if the same tilted conformation is seen using a different set of disulfides. The significance of the detergent molecule and the new cavity observed could also be better discussed in terms of an updated transport model.

<https://doi.org/10.7554/eLife.110432.1.sa1>

Author response:

We thank the three reviewers for their critical and in-depth assessment of our study. Below you find our comments to the public reviews and our revision plans.

Public Reviews:

Reviewer #1 (Public review):

This manuscript adds to the recent, exciting developments in our understanding of the MmpL/S transporters from mycobacteria. This work provides solid support for the trimeric/hexameric arrangement of subunits in the complex, and reveals a possible pathway for substrate translocation. Overall, I think this manuscript is a solid body of work that adds to several recent studies from this team and others on the structure and mechanism of the MmpL/S transporter family, particularly MmpL4/S4. The combination of AF, disulfide engineering, and experimental structure is good, though it is a bit puzzling that the experimental structure based on disulfide stabilization of the AF prediction does not recapitulate key elements (MmpS periplasmic domain docking to MmpL, and altered CCD configuration).

I have no major concerns about this manuscript; I could make only a few minor suggestions for clarification in a revision.

We thank reviewer#1 for this positive assessment of our work. The deviation of the AF prediction from the experimental structure is, in our view, not puzzling. AF does not take the physical properties of proteins into account, but predicts structures based on strong sequence alignments. It therefore does not have “knowledge” about the general flexibility of domains such as the CCD, which is also observed in the corresponding MmpL5 structures, nor does it have knowledge about preferred conformational states. Rather than “failing” to confirm the AF predictions, our cryo-EM structure revealed an unexpected tilted conformation of the CCD. As we outline in comments below, the physiological relevance of the tilted CCD is unclear. Its flexibility might be required to interact with (still elusive) outer membrane protein components to form the fully assembled efflux machinery.

Reviewer #2 (Public review):

Summary:

The manuscript describes the structure of the Mycobacterium tuberculosis (MmpS4)3-(MmpL4)3 hetero-hexameric transporter complex. The structure was obtained by cryogenic electron microscopy using an engineered construct that cross-links MmpS4 to MmpL4 via a disulfide bond. The position of the disulfide bond was determined using an Alphafold2 model of the hetero-hexamer. Although Alphafold2 predicts a symmetric hetero-hexamer, the author found that the structure of the coiled-coil domain (CCD) is asymmetric, tilted at about 60° relative to the membrane domains, and only contains two of the three alpha helical hairpins, with the third being disordered.

Strengths:

The strategy of using Alphafold2 models to guide construct design for experimental structure determination is state-of-the-art, and this work provides a great example of its applications and limitations. I.e., the experimental structure does not fully recapitulate the prediction but provides unexpected results.

The comparisons between the authors' structures and the previously published structures of the MmpL4 monomer and MmpL5 trimers strengthen the authors' findings.

We thank reviewer#2 for this positive assessment of our work and agree that it is interesting that the experimental structures do not fully agree with the AF predictions (see also comment to reviewer#1).

Weaknesses:

A more detailed description of the current mechanistic hypothesis would strengthen the manuscript. The authors state that the two periplasmic domains "are expected to undergo rigid body movements that allow substrate transport through these periplasmic domains similar to the conformational changes observed in the E. coli multidrug efflux pump AcrB". A schematic of the proposed transport cycle, as a supplemental figure that shows the current hypothesis regarding transport, would be beneficial for understanding the previous structures and putting the current structure in context. Outside of "the mechanistic basis of how these conformational changes are coupled to protonation of the DY-pairs", what are the major controversies/open questions regarding the mechanism?

We thank the reviewer for this valuable comment. We will add a new figure with the model of the MmpL4 transport cycle based on our new data and discuss the proposed molecular transport mechanism in more detail in the main.

The authors provide evidence that the cysteine-depleted S4L4 construct is functional, but do not show that the construct with the introduced disulfide bond #5 (D39C MmpS4 and S434C MmpL4) is also functional. Demonstrating this would allow the authors to better interpret their resulting structures.

In the revised version, we will include additional data to assess the functional consequences of cross-linking.

The analysis presented in Figure 5 and Supplementary Figure 7 seems to suggest that the authors are proposing that the CCD central cavity acts as a transport pathway for the transported substrate, but I am not sure that this hypothesis is explicitly stated. This makes the reasoning behind the analysis presented unclear. Clarity could be improved by stating that the hypothesis of direct transport of substrate through the CCD central channel is being examined using the structure prediction, and what the implications are for the structure solved with the incompletely formed CCD.

We state clearly in the discussion that the channel through the CCD seems too narrow to let large molecules like mycobactin and bedaquiline pass:[AG1]

Line 318ff: “ The channel radius of the MmpL4 CCD is very narrow with a minimum of 1.1 Å according to the AlphaFold3 prediction (Fig. 5). This is much smaller than the smallest axis of a molecular model of mycobactin molecule of ?? nm as determined from a model of iron-free mycobactin. In addition, the cryo-EM structure of MSMEG_1382 revealed a constriction in the CCD channel [21]. Even though the methionine side chains lining the channel wall are considered to be flexible{Aledo, 2019 #69594}, large conformational changes of the α -helical hairpins relative to each other would be required to allow passage of molecules as large as mycobactin and bedaquiline. The AcrAB-TolC efflux machinery provides an example for such large conformational changes to enable transport of large molecules by iris-like opening and closing movements the outer membrane channel-tunnel TolC [33]. Similar helical twisting may widen the channel of the CCD. Alternatively, it is conceivable that the substrates of MmpL4/MmpL5 are transported along the CCD surface, potentially requiring further protein partners. It is interesting to note that siderophore secretion and drug efflux by MmpL4/MmpL5 systems involves at least two additional proteins, namely the periplasmic protein Rv0455, which was shown to be essential for mycobactin efflux [34] and an outer membrane channel, whose identity remains elusive. A complete molecular understanding of the transport mechanism through the MmpL4/MmpL5 systems hence requires the identification of the missing components and structural information about their interactions.”

The channel radius of the MmpL4 CCD is very narrow (minimum of 1.1 Å) according to the AlphaFold3 prediction (Fig. 5), and the cryo-EM [AG2] [MN3] structure of MSMEG_1382 revealed a further constriction in the CCD channel [21]. We therefore consider direct substrate transport through the CCD central channel to be physically implausible for molecules of the size of mycobactin and bedaquiline. Even accounting for the flexibility of the methionine side chains lining the channel wall, the large conformational changes of the α -helical hairpins relative to each other would be required to accommodate such large substrates. While iris-like opening movements have been described for TolC in the AcrAB-TolC system [33], those movements widen an already substantially larger channel, and even such dramatic conformational changes would be insufficient to open a channel as narrow as that of the MmpL4 CCD to a diameter permissive for substrate passage. We instead favor a model in which substrates are transported along the outer surface of the CCD, potentially with the assistance of additional protein partners. This is consistent with the observation that MmpL4/MmpL5-mediated siderophore secretion and drug efflux involves at least two further proteins: the periplasmic protein Rv0455, shown to be essential for mycobactin efflux [34], and an as-yet-unidentified outer membrane channel. In this context, the overall flexibility of the CCD - illustrated here by the tilted, incompletely formed conformation - may reflect the conformational dynamics required for interaction with these partner proteins, rather than being directly involved in forming a transport conduit. A complete mechanistic understanding will require identification of the missing components and structural characterization of the fully assembled efflux machinery.

We do not think that the incompletely formed CCD represents a conformation that is relevant for transport. But it is a demonstration of the overall flexibility of the CCD, which may be required to further open the channel in case the substrates are transported within the CCD tube. Further in-depth experiments will be needed to clarify this interesting question, which is beyond the scope of this paper.

Given that the results emphasize the flexibility of the CCD, the manuscript would be strengthened by 3D variability analysis either in cryoSPARC or using cryoDRGN (or both). This would allow the authors to better quantify the degree of motion in the CCD and how

it may correlate to flexibility in other regions. Further 3D flex reconstruction in cryoSPARC may improve the map quality of the CCD.

This is a great suggestion. We will include a 3D variability analysis in the revised manuscript.

Reviewer #3 (Public review):

Summary:

This manuscript by Earp et al reports cryoEM structures of the hexameric (MmpS4)₃-(MmpL4)₃ complex from Mycobacterium tuberculosis, which belongs to the RND family of transporters and is known to have a role in the export of siderophores and contribute to drug resistance. The experimental workflow showcased involves the design of disulfide pairs using distance constraints obtained from the AlphaFold predicted structure of the hexameric complex. One such disulfide pair was used to determine the ~3.0 Å structures. The structure reveals density for the previously unresolved coiled-coil domain (CCD), a tilted CCD arrangement, and a cavity within the periplasmic domain, which the authors assert is occupied by detergent. Comparison of this complex with the monomer structure of MmpL4 shows conformational variations interpreted to implicate different domains and conserved residues involved in proton coupling, which might be related to the transport mechanism. While the methodological aspects of the manuscript are solid, enthusiasm for the overall advance/significance is less so, with doubts about the relevance of the tilted CCD structure, considering disulfide trapping and an incomplete validation of the claim that the tilted CCD represents a stable intermediate conformation. A clear, updated transport mechanism is largely missing from the manuscript.

We thank reviewer#3 for these useful comments, which we will address during the revision of the manuscript. In particular, we plan to include a scheme of an updated transport model.

Strengths:

Beautiful structures, AF prediction-experimental validation nexus that could be fine-tuned for different systems/difficult to target complexes.

Weaknesses:

Physiological relevance of the tilted CCD conformation. No clear mechanistic model for the transport. While the CCD may indeed be a stable intermediate, the fact that the rest of the trimeric arrangement is unaffected does not fully rule out disulfide trapping as a factor in promoting this. The findings would be strengthened if the same tilted conformation is seen using a different set of disulfides. The significance of the detergent molecule and the new cavity observed could also be better discussed in terms of an updated transport model.

We believe that there was a misunderstanding about our interpretation of the tilted CCD. As a matter of fact, it must be a stable intermediate, otherwise no density would have been observed for it in the cryo-EM maps. Despite being a stable intermediate, it is indeed unlikely that it represents a conformational state that is relevant/required for transport. Firstly, only the upright, complete CCD can bridge the periplasm. because . Secondly, the structure was determined in detergent and lacks additional protein binder partners, which might stabilize the upright conformation of the CCD . It is also conceivable, as the reviewer pointed out, that disulfide cross-linking may have caused the tilt. However, as we wrote in the manuscript, we do not think that cross-linking caused this striking asymmetry of the CCD, because the three MmpL4 and MmpS4 chains are basically symmetrical in the C1-processed data (see also Figure 2E):

Line 182 ff: “To assess whether there are asymmetries in other parts of the structure, we superimposed the individual protomers of the (MmpS4)₃-(MmpL4)₃ complex analyzed using C1 symmetry (Fig. 2E). Apart from the two resolved α -helical hairpins, the MmpL4 core domains and the resolved parts of MmpS4 differ by a RMSD of less than 0.6 Å and are therefore structurally identical considering the map resolution of around 3 Å. The fact that the core domains of MmpS4 and MmpL4 do not deviate between the protomers argues against the possibility that the cross-links established between them cause the (asymmetric) tilt of the CCD.”

Regarding the DDM binding site, we will indeed include an updated transport model. That said, we wish to be cautious, because we lack experimental proof that MmpL4 can in fact transport DDM.

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